

Real-Time Time-Dependent Density Functional Theory for Periodic and Spin-Polarized Systems with the Real-Space Multigrids (RMG)[†]

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Abstract

[Abstract: to be written last, after all sections are complete. Approximately 200 words. Should mirror paper 1 abstract structure: (1) what we present, (2) what problem it solves, (3) key methodological contribution, (4) validation/benchmarking, (5) scalability demonstration and outlook.]

1 Introduction

Real-time time-dependent density functional theory (RT-TDDFT) has established itself as one of the primary tools for simulating the electronic response of materials to external electromagnetic perturbations.^{??} Its applications span a broad range of phenomena, including optical spectroscopy,^{????} charge transfer and photovoltaics,^{???} nanoplasmonics,^{???} and spin and magnetization dynamics.^{???} In an earlier publication, we presented a scalable RT-TDDFT implementation within the Real-space MultiGrid (RMG) code for finite molecular systems, demonstrating simulations of up to 24,000 electrons on massively parallel architectures.[?] The present work extends that implementation to periodic systems and spin-polarized calculations, significantly broadening the range of physically relevant problems accessible to RMG.

The transition from finite to periodic systems in RT-TDDFT requires a corresponding transition in the treatment of the electromagnetic perturbation. In finite systems, the length

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gauge—in which the coupling to an electric field is expressed through the dipole operator $\hat{\mathbf{r}}$ —is natural and convenient. In periodic systems, however, the position operator is ill-defined because it is not translationally invariant, and the dipole moment of an extended system is intrinsically multivalued.^{??} The correct framework for periodic systems is the velocity gauge, in which the electromagnetic interaction is expressed through the vector potential $\mathbf{A}(t)$ and the current operator, which is inherently translationally invariant. The transition between the two gauges is effected by a gauge transformation of the time-dependent Kohn-Sham equations,^{???} and requires careful treatment of the nonlocal part of pseudopotentials.^{???} The primary observable in the velocity gauge is the macroscopic current density $\mathbf{J}(t)$, from which the optical conductivity $\sigma(\omega)$ and dielectric function $\epsilon(\omega)$ are extracted by Fourier analysis.

RT-TDDFT for periodic systems has been implemented in several codes. The `Octopus` code,[?] based on real-space grids, provides a comprehensive periodic RT-TDDFT implementation including both length and velocity gauges and has served as the primary validation reference for the present work. Plane-wave codes including `Qbox/Qball`,^{??} `cp2k`,[?] and `Exciting`[?] offer periodic RT-TDDFT with natural handling of Born-von Kármán boundary conditions. Atom-centered basis set codes such as `SIESTA`[?] and `DFTB+`[?] provide efficient implementations for systems where localized orbitals are appropriate. The `DFT-FE` code[?] offers an alternative real-space approach based on finite elements. [\[Check: add any missing periodic RT-TDDFT codes before submission. In particular: GPAW, Abinit, Yambo.\]](#)

The treatment of spin-polarized systems within RT-TDDFT introduces an additional degree of freedom: the electronic density is resolved into spin-up and spin-down components, and the time-dependent exchange-correlation potential depends on both. In the collinear approximation, the two spin channels evolve independently subject to a common self-consistent spin-dependent potential, and the optical response carries spin-specific signatures. This capability is essential for magnetic insulators such as CrI_3 , where the spin-up and spin-down optical spectra differ markedly and encode information about the electronic and magnetic structure.^{??} The non-collinear spin extension, in which the electronic states are represented as two-component spinors, is beyond the scope of the present work and will be addressed in a future publication.

A distinctive feature of the `RMG` implementation is its scalability. The real-space multi-grid approach assigns different regions of space to different processors, enabling near-linear scaling to hundreds or thousands of GPU-accelerated nodes.^{??} This scalability is particularly significant for periodic calculations with spin, where both the k -point sampling and the spin-polarized density matrix increase the computational cost relative to finite systems. To demonstrate this capability, we apply the periodic spin-polarized RT-TDDFT implementation to nitrogen-vacancy (NV) centers in diamond—a prototypical point defect system of substantial interest for quantum sensing and quantum information applications.[?] By systematically increasing the supercell size from [\[X\]](#) to [\[Y\]](#) atoms, we show how the optical response of the defect converges as the interaction between periodic images becomes negligible, identifying the minimum supercell size required for physically meaningful results at realistic defect concentrations.

The remainder of this paper is organized as follows. Section 8 develops the theoretical framework, beginning with the interaction of matter with electromagnetic radiation in the long-wavelength approximation, proceeding through the gauge transformation from length to velocity gauge, the definition of the current operator including nonlocal pseudopoten-

tial contributions, the density matrix propagation scheme, the spin-collinear extension, and the postprocessing route from current to optical conductivity. Section 3 describes the algorithm as implemented in **RMG** and identifies the key approximations and their physical justification. Section 5 presents benchmarking calculations for one-dimensional (H_2 chain), two-dimensional (hexagonal BN), and three-dimensional (Si) periodic systems, followed by spin-polarized calculations for CrI_3 , with comparisons to **Octopus** in all cases. The section concludes with the NV-center in diamond scalability demonstration. The paper concludes with a Summary and Outlook.

2 Methodology

This section presents the theoretical foundations of real-time time-dependent density functional theory (RT-TDDFT) for periodic systems interacting with electromagnetic radiation. We begin with the working equations for the electromagnetic field and the minimal coupling Hamiltonian (Section 2.1), then develop the gauge transformation from the length gauge to the velocity gauge (Section 2.2), the electronic current density operator including nonlocal pseudopotential contributions (Section 2.3), and the time propagation of the Kohn-Sham equations (Section 2.4). Rather than stating final expressions alone, we provide complete, step-by-step derivations with all intermediate results and sign conventions made explicit. Our motivation for this level of detail is practical: the existing literature on velocity-gauge RT-TDDFT is spread across several communities - solid-state physics, quantum chemistry, and optical response theory - each employing different conventions for the electron charge sign, the minimal coupling substitution, and the definition of the current operator. Reconstructing a consistent set of working equations from these sources requires reconciling notational conflicts and filling in steps that are frequently omitted. We aim to present a self-contained derivation that can serve as a reference for implementation. To this end, we work in Gaussian units throughout the theory section, keeping $\hbar$, m , and e explicit so that physical dimensions and charge signs remain traceable at every step. Atomic units ($\hbar = m_e = e = 4\pi\epsilon_0 = 1$) are adopted starting in Section 3 for the working equations as implemented in the code.

2.1 Interaction of Matter with Electromagnetic Radiation

The starting point for any RT-TDDFT calculation is the time-dependent Kohn-Sham equation for single-particle orbitals $\Psi_j(\mathbf{r}, t)$ in the presence of external electromagnetic fields which, with the minimal coupling framework, takes the form

$$i\hbar \frac{\partial}{\partial t} \Psi_j(\mathbf{r}, t) = \left[\frac{1}{2m} \left(\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}_{\text{ext}}(\mathbf{r}, t) \right)^2 + q_e \Phi_{\text{ext}}(\mathbf{r}, t) + \hat{V}_{\text{KS}}[\rho](t) \right] \Psi_j(\mathbf{r}, t) \quad (1)$$

where $\mathbf{A}_{\text{ext}}$ and Φ_{ext} are vector and scalar potential representing external electromagnetic (EM) field, $\hat{\mathbf{p}} = -i\hbar\nabla$ is a canonical momentum operator, and $q_e = -1$ is an electron charge. The Kohn-Sham potential, $\hat{V}_{\text{KS}}[\rho](t)$, contains the ionic potential, Hartree potential,

and exchange-correlation potential:

$$\hat{V}_{\text{KS}} = \hat{V}_{\text{ion}} + \hat{V}_{\text{H}}[\rho](\mathbf{r}, t) + \hat{V}_{\text{xc}}[\rho](\mathbf{r}, t). \quad (2)$$

In practical implementations based on norm-conserving or ultrasoft pseudopotentials, $\hat{V}_{\text{ion}}$ may contain both local and nonlocal parts, and gauge transformations of the Hamiltonian may introduce additional terms that must be accounted for in both the Hamiltonian and the current density operator. The choice of the specific form of $\mathbf{A}_{\text{ext}}$ and Φ_{ext} depends on the problem and is subject to a gauge freedom.[?]

For a monochromatic external EM field, the vector potential may be represented as a plane wave,

$$\mathbf{A}_{\text{ext}}(r, t) = \mathbf{A}_0 e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} + c.c. \quad (3)$$

where ω is the angular frequency, a wave vector $\mathbf{q}$ describes propagation direction, and $\mathbf{A}_0 = [A_x^0, A_y^0, A_z^0]$ is a complex polarization amplitude. The corresponding electric and magnetic fields are obtained from Maxwell equations:

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= -\nabla \Phi_{\text{ext}}(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathbf{A}_{\text{ext}}(\mathbf{r}, t)}{\partial t}, \\ \mathbf{B}(\mathbf{r}, t) &= \nabla \times \mathbf{A}_{\text{ext}}(\mathbf{r}, t). \end{aligned}$$

Any vector field $\mathbf{A}(\mathbf{r}, t)$ can be uniquely decomposed into a *transverse* (divergence-free) component perpendicular to $\mathbf{q}$ and a *longitudinal* (curl-free) component parallel to $\mathbf{q}$. In the Coulomb gauge framework, defined by the condition

$$\nabla \cdot \mathbf{A}_{\text{ext}}(\mathbf{r}, t) = 0, \quad (4)$$

the vector potential is purely transverse: it carries only the transverse part of the electric field and the magnetic field, while the scalar potential Φ_{ext} carries the longitudinal part of the electric field exclusively. For a plane wave, this becomes apparent upon applying the gauge condition (4) to Eq. (3), which yields the transversality condition

$$\mathbf{q} \cdot \mathbf{A}_0 = 0, \quad (5)$$

requiring that the polarization of the radiation field is perpendicular to its propagation direction.

In general, the electromagnetic potentials ($\mathbf{A}$ and Φ) and the KS potential all appear simultaneously. However, the gauge freedom allows to transform $\mathbf{A}$ and Φ to form suitable for a given problem using an arbitrarily selected scalar function $\chi(r, t)$ and a set of transformation rules

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla \chi(r, t), \quad (6)$$

$$\Phi \rightarrow \Phi - \frac{1}{c} \frac{\partial \chi(r, t)}{\partial t}, \quad (7)$$

A specific choice of χ relevant to periodic systems is discussed in Sec. 2.3.

In chemistry applications, the most common choice for RT-TDDFT is the *length gauge*, in

which $\mathbf{A} = 0$ and the external electromagnetic field enters entirely through a magnetic-field-free scalar potential. In the long-wavelength limit, applicable when the size of the system L satisfies $|\mathbf{q}|L \ll 1$ (i.e. L is much smaller than the radiation wavelength $\lambda = 2\pi/|\mathbf{q}|$), the external field is effectively spatially uniform over the system and the interaction reduces to the electric-dipole approximation. A spatially uniform electric field $\mathbf{E}(t)$ is represented by the dipolar scalar potential

$$\Phi(\mathbf{r}, t) = -\mathbf{E}(t) \cdot \mathbf{r}, \quad (8)$$

as can be verified from $\mathbf{E} = -\nabla\Phi$. This form is widely used to investigate the optical response of finite molecular systems.

While the plane-wave form (3) describes a monochromatic perturbation, in practice the time-oscillating factor may be replaced by an arbitrary envelope function $f(t)$, so that the applied electric field takes the form $\mathbf{E}(t) = \hat{\mathbf{e}} E_0 f(t)$, where $\hat{\mathbf{e}}$ is the polarization direction. A particularly useful choice is the Dirac delta kick, $f(t) = \kappa \delta(t)$, which delivers a broadband impulsive perturbation of strength κ at $t = 0$. Since the delta function has a flat spectral content, a single time propagation yields the dynamical response at all frequencies simultaneously. The frequency-dependent absorption spectrum is then obtained from the Fourier transform of the time-dependent observable monitored during the propagation.

For finite systems, the relevant observable is the time-dependent dipole moment $\mathbf{d}(t) = \int \mathbf{r} \rho(\mathbf{r}, t) d\mathbf{r}^3$, from which the polarizability and absorption cross section are obtained from a dipole correlation function by Fourier transform. For periodic systems, however, the scalar-potential form $\Phi = -\mathbf{E}(t) \cdot \mathbf{r}$ is not compatible with periodic boundary conditions, because the position operator $\mathbf{r}$ is discontinuous at the cell boundary. This motivates the use of the velocity-gauge formulation, in which the external field enters through a spatially uniform vector potential $\mathbf{A}(t)$ and the scalar potential is set to zero. For finite systems, the optical response is extracted from the time-dependent dipole moment, which depends on the density, whereas for truly periodic systems the relevant observable is the current density $\mathbf{j}(\mathbf{r}, t)$, which remains well defined under periodic boundary conditions.

In a summary, it is useful at this point to recognize that external perturbations in RT-TDDFT fall into two physically distinct categories. The first category comprises electromagnetic field perturbations, in which the external field enters through the vector potential in the kinetic energy via the minimal coupling $\hat{\mathbf{p}} \rightarrow \hat{\mathbf{p}} - (q_e/c)\mathbf{A}$. The choice of gauge (length or velocity) determines how this coupling is represented in the Hamiltonian, and the long-wavelength limit leads to the dipolar approximation discussed above. Typical applications include optical absorption spectra and laser-driven electron dynamics.

The second category comprises scalar potential perturbations of the form

$$\Phi_{\text{ext}}(\mathbf{r}, t) = \Phi_0 e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} + \text{c.c.}, \quad (9)$$

which are added directly to the Kohn–Sham potential $\hat{V}_{\text{KS}}$ and involve no gauge freedom. The electric field generated by such a potential, $\mathbf{E} = -\nabla\Phi_{\text{ext}} \propto i\mathbf{q} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$, is purely longitudinal (parallel to $\mathbf{q}$) and carries no magnetic field. This type of perturbation probes the density-density response function $\chi(\mathbf{q}, \omega)$ and is the natural tool for studying collective excitations at finite momentum transfer, such as plasmon dispersion and charge-density oscillations.

Although both categories may involve the same spatial factor $e^{i\mathbf{q} \cdot \mathbf{r}}$, they are physically

Table 1: Summary of common external-field representations used in TDDFT, applicable to both linear-response and real-time formulations. In Coulomb gauge, the scalar potential carries the longitudinal part of the electric field, while the transverse vector potential carries the transverse electric field and the magnetic field.

Case		Potentials	Field character	B field	Typical applications
Length dipole	gauge,	$\Phi = -\mathbf{r} \cdot \mathbf{E}(t), \mathbf{A} = 0$	homogeneous electric field	no	optical absorption of finite systems, photochemistry ^{??}
Velocity	gauge	$\Phi = 0, \mathbf{A} = \mathbf{A}(t)$	homogeneous field	no	optical response of solids and periodic systems under PBC ^{???}
Finite- $\mathbf{q}$ probe	scalar	$\Phi \propto e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}, \mathbf{A} = 0$	longitudinal, $\mathbf{E} \parallel \mathbf{q}$	no	density response, loss function, EELS, plasmon dispersion ^{???}
Finite- $\mathbf{q}$ wave	EM	$\mathbf{A} \propto \boldsymbol{\epsilon} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}, \mathbf{q} \cdot \boldsymbol{\epsilon} = 0$	transverse EM radiation	yes	IXS, radiative coupling beyond dipole, magnon dispersion ^{??}

and computationally distinct: the electromagnetic perturbation couples to the electron momentum through the kinetic energy and requires gauge treatment, while the scalar perturbation couples to the electron density through the potential energy and enters the calculation without any gauge considerations.

Many important spectroscopies and collective excitations involve finite momentum transfer and lie beyond the dipolar approximation. These include plasmon dispersion $\varepsilon(\mathbf{q}, \omega)$, electron energy-loss spectroscopy (EELS) where the beam transfers finite $\mathbf{q}$ to the sample, inelastic X-ray scattering (IXS), and magnon dispersion in magnetic systems. Table 1 summarizes the common external-field representations used in TDDFT and their typical applications.

2.2 VELOCITY GAUGE

For finite systems, the relevant observable is the time-dependent dipole moment $\mathbf{d}(t) = \int \mathbf{r} \rho(\mathbf{r}, t) d\mathbf{r}$, from which the polarizability and absorption cross section follow by Fourier transform. For periodic systems, however, the scalar-potential form $\Phi = -\mathbf{E}(t) \cdot \mathbf{r}$ is not compatible with periodic boundary conditions, because the position operator $\mathbf{r}$ is discontinuous at the cell boundary. This motivates the use of the velocity-gauge formulation, in which the external field enters through a spatially uniform vector potential $\mathbf{A}(t)$ and the scalar potential is set to zero.

The time-dependent Kohn–Sham equation then takes the form

$$i \frac{\partial}{\partial t} \Psi_j(\mathbf{r}, t) = \left[\frac{1}{2m} \left(\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(t) \right)^2 + \hat{V}_{\text{KS}}[\rho](t) \right] \Psi_j(\mathbf{r}, t), \quad (10)$$

where the two gauges are related by the Göppert-Mayer transformation, discussed in detail in Sec. 2.3. In the velocity gauge, the position operator no longer appears in the Hamil-

tonian and all terms are compatible with periodic boundary conditions. Correspondingly, the physical observable used to extract the optical response is the time-dependent current density $\mathbf{j}(\mathbf{r}, t)$ rather than the dipole moment, since the current is well defined under periodic boundary conditions whereas $\mathbf{d}(t)$ is not.

2.3 Gauge transformations

In this section we discuss transformation of Time-Dependent Kohn-Sham equation under different gauge representation within Coulomb gauge condition. The starting point are general Eqs. [A' $\rightarrow$ A + ' ; Phi' $\rightarrow$ Phi +] and the requirement that physically measurable quantities such mechanical momentum, velocities and physically measurable operator observable remain unchanged. The invariance of E and B is directly guaranteed by [XXX]

For finite systems, the relevant observable is the time-dependent dipole moment $\mathbf{d}(t) = \int \mathbf{r} \rho(\mathbf{r}, t) d\mathbf{r}$, from which the polarizability and absorption cross section follow by Fourier transform. For periodic systems, however, the scalar-potential form $\Phi = -\mathbf{E}(t) \cdot \mathbf{r}$ is not compatible with periodic boundary conditions, because the position operator $\mathbf{r}$ is discontinuous at the cell boundary. This motivates the use of the velocity-gauge formulation, in which the external field enters through a spatially uniform vector potential $\mathbf{A}(t)$ and the scalar potential is set to zero.

The time-dependent Kohn-Sham equation then takes the form

$$i\frac{\partial}{\partial t}\Psi_j(\mathbf{r}, t) = \left[\frac{1}{2m} \left(\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(t) \right)^2 + \hat{V}_{\text{KS}}[\rho](t) \right] \Psi_j(\mathbf{r}, t), \quad (11)$$

where the two gauges are related by the Göppert-Mayer transformation, discussed in detail in Sec. 2.3. In the velocity gauge, the position operator no longer appears in the Hamiltonian and all terms are compatible with periodic boundary conditions. Correspondingly, the physical observable used to extract the optical response is the time-dependent current density $\mathbf{j}(\mathbf{r}, t)$ rather than the dipole moment, since the current is well defined under periodic boundary conditions whereas $\mathbf{d}(t)$ is not.

3 Implementation

The implementation of the periodic and spin-polarized RT-TDDFT presented in this work is built on the existing finite-system RT-TDDFT module of **RMG**,[?] which is described in detail in the accompanying paper. Here we focus on the extensions required for periodic boundary conditions and spin-polarized calculations, the key approximations and their physical justification, and the computational organization of the velocity gauge propagation.

3.1 Algorithm

The computational scheme for periodic RT-TDDFT in **RMG** is summarized in Algorithm 1. It follows the same density matrix propagation framework as the finite-system implementation[?] but with three key modifications: (1) the perturbation is applied as a momentum kick via the vector potential $\mathbf{A}$ rather than an electric field dipole kick; (2) the current operator

matrix replaces the dipole matrix as the primary observable; and (3) the Hamiltonian matrix is evaluated at each k-point over the Brillouin zone.

Algorithm 1 Periodic RT-TDDFT Algorithm (VECTOR_POT mode)

```

/* Ground state DFT calculations: */
1: Generate Bloch orbitals  $|\psi_{n,\mathbf{k}}\rangle$ , Kohn-Sham Hamiltonian  $\mathbf{H}_{0,\mathbf{k}}^\sigma$  for each k-point  $\mathbf{k}$  and spin  $\sigma$ 
2: Select active space:  $N_{\text{occ}}$  occupied +  $N_{\text{virt}}$  virtual orbitals

/* Build current operator matrices (once, before time loop): */
3: For each k-point  $\mathbf{k}$  and spin  $\sigma$ :
4:   Build paramagnetic current matrix:
      $J_{ab}^\alpha \leftarrow \langle \psi_a | (-i\nabla_\alpha + k_\alpha) | \psi_b \rangle$  [VecPHmatrix()]
5:   Add nonlocal PP correction:
      $J_{ab}^\alpha \leftarrow J_{ab}^\alpha + \langle \psi_a | [\hat{r}_\alpha, V^{\text{nl}}] | \psi_b \rangle$  [CurrentNlpp()]
6:   Add  $\mathbf{A}(t_0) \cdot \mathbf{J}$  to  $\mathbf{H}_{-1,\mathbf{k}}$  to encode the perturbation kick at  $t = 0$ 

/* Initialize density matrix: */
7:  $\mathbf{P}_\mathbf{k}^0 \leftarrow \text{diag}(f_{n,\mathbf{k}})$  (occupation numbers on diagonal)
8:  $\mathbf{H}_{-1,\mathbf{k}} \leftarrow \mathbf{H}_{0,\mathbf{k}}$  (extrapolation seed)

/* Time loop: */
9: for  $j = 1$  to  $N_{\text{steps}}$  do
10:    $t \leftarrow t_0 + j \cdot \Delta t$ 
11:   Extrapolate:  $\mathbf{H}_{1,\mathbf{k}} \leftarrow 2\mathbf{H}_{0,\mathbf{k}} - \mathbf{H}_{-1,\mathbf{k}}$  [extrapolate_Hmatrix()]

/* Self-consistent Hamiltonian loop: */
12:   while  $\|\mathbf{H}_1^{\text{new}} - \mathbf{H}_1^{\text{old}}\|_\infty > H_{\text{thrs}}$  do
13:     Magnus operator:  $\Omega_\mathbf{k} \leftarrow -i \cdot \frac{1}{2} (\mathbf{H}_{0,\mathbf{k}} + \mathbf{H}_{1,\mathbf{k}}) \Delta t$  [magnus()]
14:     Propagate density matrix:  $\mathbf{P}_\mathbf{k}^1 \leftarrow \mathbf{P}_\mathbf{k}^0 + [\Omega_\mathbf{k}, \mathbf{P}_\mathbf{k}^0] + \dots$  [eldyn_ort() / commutp()]
15:     Update density:  $\rho^\sigma(\mathbf{r}) \leftarrow \rho_{\text{gs}}^\sigma(\mathbf{r}) + \sum_\mathbf{k} w_\mathbf{k} \sum_n P_{nn,\mathbf{k}}^1 |\psi_{n,\mathbf{k}}(\mathbf{r})|^2$  [GetNewRho_rmgtddft()]
16:     Update potentials:  $V_{\text{H}}[\rho], V_{\text{xc}}^\sigma[\rho^\uparrow, \rho^\downarrow]$ 
17:     Update Hamiltonian:  $\mathbf{H}_{1,\mathbf{k}}^{\text{new}} \leftarrow \langle \psi_a | \hat{H}[\rho] | \psi_b \rangle$  [HmatrixUpdate()]
18:   end while

19: Extract current:  $J_\alpha(t) \leftarrow \text{Re} \sum_\mathbf{k} w_\mathbf{k} \text{Tr}[\mathbf{P}_\mathbf{k}^1 \cdot \mathbf{J}_\mathbf{k}^\alpha]$ 
20: (Optional) Berry phase polarization:  $P_{\text{BP},\alpha}(t) \leftarrow \text{Tr}[\mathbf{P}_\mathbf{k}^1 \cdot \mathbf{X}_\mathbf{k}^\alpha]$ 
21: Advance:  $\mathbf{P}^0 \leftarrow \mathbf{P}^1$ ;  $\mathbf{H}_{-1} \leftarrow \mathbf{H}_0$ ;  $\mathbf{H}_0 \leftarrow \mathbf{H}_1$ 
22: end for

/* Postprocessing: */
23: Fourier transform  $J_\alpha(t)$  for each Cartesian direction  $\alpha$ 
24: Compute  $\sigma_{\alpha\beta}(\omega), \varepsilon_{\alpha\beta}(\omega)$ 

```

3.2 Key Approximations and Their Justification

The implementation described in Algorithm 1 embodies several approximations with respect to the full theory presented in Section 8. We discuss each in turn.

Perturbative momentum kick. The full velocity gauge Hamiltonian (eq. 40) contains the vector potential $\mathbf{A}(t)$ at every time step, requiring the $\mathbf{A}(t) \cdot \hat{\mathbf{p}}$ term to be re-evaluated throughout the propagation. In the present implementation, we instead apply $\mathbf{A}(t)$ as a delta-function kick at $t = 0$ only: the momentum kick is encoded in the initial Hamiltonian $\mathbf{H}_{-1}$ by adding the $\mathbf{A}(t_0) \cdot \mathbf{J}^\alpha$ matrix (step 6 in Algorithm 1), after which $\mathbf{A} = 0$ for all $t > 0$. This approach is equivalent to first-order perturbation theory in $|\mathbf{A}_0|$, and it provides the full linear optical response of the system through the free oscillations of the current density $\mathbf{J}(t)$ following the kick. The linear response regime is valid as long as the kick amplitude is

small enough that the system does not leave the linear regime, which is straightforwardly verified by checking that the response is proportional to $|\mathbf{A}_0|$.

Diamagnetic current term. The full current density contains both paramagnetic and diamagnetic contributions (Section 8.5). In the present implementation, the diamagnetic term $\mathbf{J}_{\text{dia}} \propto \mathbf{A}(t)|\psi|^2$ is identically zero for all $t > 0$ because $\mathbf{A}(t > 0) = 0$ in the perturbative kick scheme. This is consistent with the linear response regime: the diamagnetic contribution to the linear response function is implicitly accounted for by the sum rule, and the paramagnetic current alone gives the correct optical conductivity.^{??} Simulations with a continuously applied driving field (e.g., a CW laser or a Gaussian pulse) would require retaining $\mathbf{A}(t)$ throughout the propagation and are beyond the scope of the present work.

First-order Magnus expansion. As in the finite-system implementation,[?] the Magnus operator is truncated to first order:

$$\Omega \approx \Omega_1 = -i \cdot \frac{1}{2}(\mathbf{H}_0 + \mathbf{H}_1)\Delta t. \quad (12)$$

The second-order correction $\Omega_2 = \frac{1}{12}\Delta t^2[\mathbf{H}_0, \mathbf{H}_1]$ is not included. The accuracy and stability of this scheme for the time steps used here was validated in paper 1 through energy conservation tests, and we demonstrate analogous stability for the periodic case in Section 5.

Norm-conserving pseudopotentials only. The current implementation supports only norm-conserving pseudopotentials (NCPP).^{??} Ultrasoft pseudopotentials (USPP)[?] introduce a non-trivial overlap matrix $\mathbf{S} \neq \mathbf{I}$ into the density matrix propagation (the nonorthogonal basis path), which requires the generalized propagation scheme described in Appendix A of paper 1. This extension is planned for future work.

BCH propagator for complex Hamiltonians. For k-points away from the Γ point, the Hamiltonian matrix $\mathbf{H}_{\mathbf{k}}$ is complex Hermitian. In this case, the BCH commutator expansion propagator (Ieldyn=1) is used throughout; the diagonalization-based propagator (Ieldyn=2) is not currently supported for complex matrices. The BCH expansion is unconditionally stable and has been shown to give results equivalent to exact diagonalization within the convergence threshold,[?] and we find no practical limitation from this choice.

3.3 Software Optimization and Parallelization

[Content outline – adapt from paper 1 Section 3.2 with] additions for the periodic case:
- Same MPI + OpenMP + GPU parallelization as paper 1 - k-point parallelization: each MPI group handles a subset of k-points (pct.kpsub_comm); current is summed over k-points with weights (MPI_Allreduce over kpsub_comm, lines 828-829 in RmgTddft.cpp) - Spin parallelization: separate MPI communicator groups per spin channel (pct.spin_comm); spin channels communicate only for XC update (MPI_Allreduce over spin_comm, line 767)
- The density matrix P and Hamiltonian H are distributed using ScaLAPACK layout (Sp object); size is numst x numst where numst = N_occ + N_virt - For VECTOR_POT mode:

momentum matrix J_{α} is complex even at Gamma point (stored as complex<double> in `Pxmatrix_cpu` etc.) - The BCH commutator $[\Omega, P]$ dominates $O(N^3)$; same optimization as paper 1 (real/imaginary decomposition, BLAS ZGEMM/DGEMM) - Timing data for benchmark systems: [add when available]

[Note: include a brief timing comparison table analogous to Table 1 in paper 1, but for one or more periodic benchmark systems (e.g., Si supercells of increasing size) to demonstrate scaling.]

4 Numerical Results

We benchmark the periodic and spin-polarized RT-TDDFT implementation in **RMG** against the **Octopus** code[?] for four representative systems of increasing dimensionality and physical complexity: a one-dimensional hydrogen chain (H_2), a two-dimensional hexagonal boron nitride (hBN) monolayer, a three-dimensional silicon crystal, and the spin-polarized magnetic insulator CrI_3 . These systems are chosen to isolate and validate specific aspects of the implementation: k-point sampling, anisotropic optical response, three-dimensional periodic boundaries, and spin-collinear propagation, respectively. The section concludes with a demonstration of the unique large-scale capability of **RMG** through calculations on nitrogen-vacancy (NV) centers in diamond at varying supercell sizes.

In all calculations, a momentum kick of amplitude $E_0 = [\text{value}]$ a.u. was applied at $t = 0$ along each Cartesian direction independently, and the optical conductivity was obtained from the Fourier transform of the resulting current density. [Add: common computational parameters – time step, total propagation time, broadening gamma, XC functional, pseudopotential library used for all systems.]

4.1 One-Dimensional Periodic System: Hydrogen Chain

As the simplest possible periodic test system, we consider an infinite chain of hydrogen molecules aligned along the z axis with alternating bond lengths, modeled by a one-dimensional supercell with periodic boundary conditions. This system provides a stringent test of k-point sampling and the velocity gauge implementation in the limit of a quasi-one-dimensional geometry, where the longitudinal (z -direction) and transverse (x, y -direction) optical responses differ substantially.

[Computational details: - Supercell geometry: H–H bond length, H_2 – H_2 separation - K-point sampling: $[N_k]$ points along z , Gamma in x and y - Grid spacing: $[\text{value}] a_0$ - XC functional: [PBE or LDA] - Pseudopotential: [type] - Time step Δt : $[\text{value}]$ a.u. - Total simulation time: $[\text{value}]$ a.u. - Broadening γ : $[\text{value}]$ eV]

[Discussion points: - Agreement between **RMG** and **Octopus** for both longitudinal and transverse components - K-point convergence: how many k-points needed for converged spectrum? - Physical interpretation of the spectral features - If Berry phase comparison is available: show agreement between current-integration and BP polarization as internal consistency check (Fruit 1 from CONTEXT.md) - If ground-state current convergence is shown: include as panel or note in text (Fruit 2 from CONTEXT.md)]

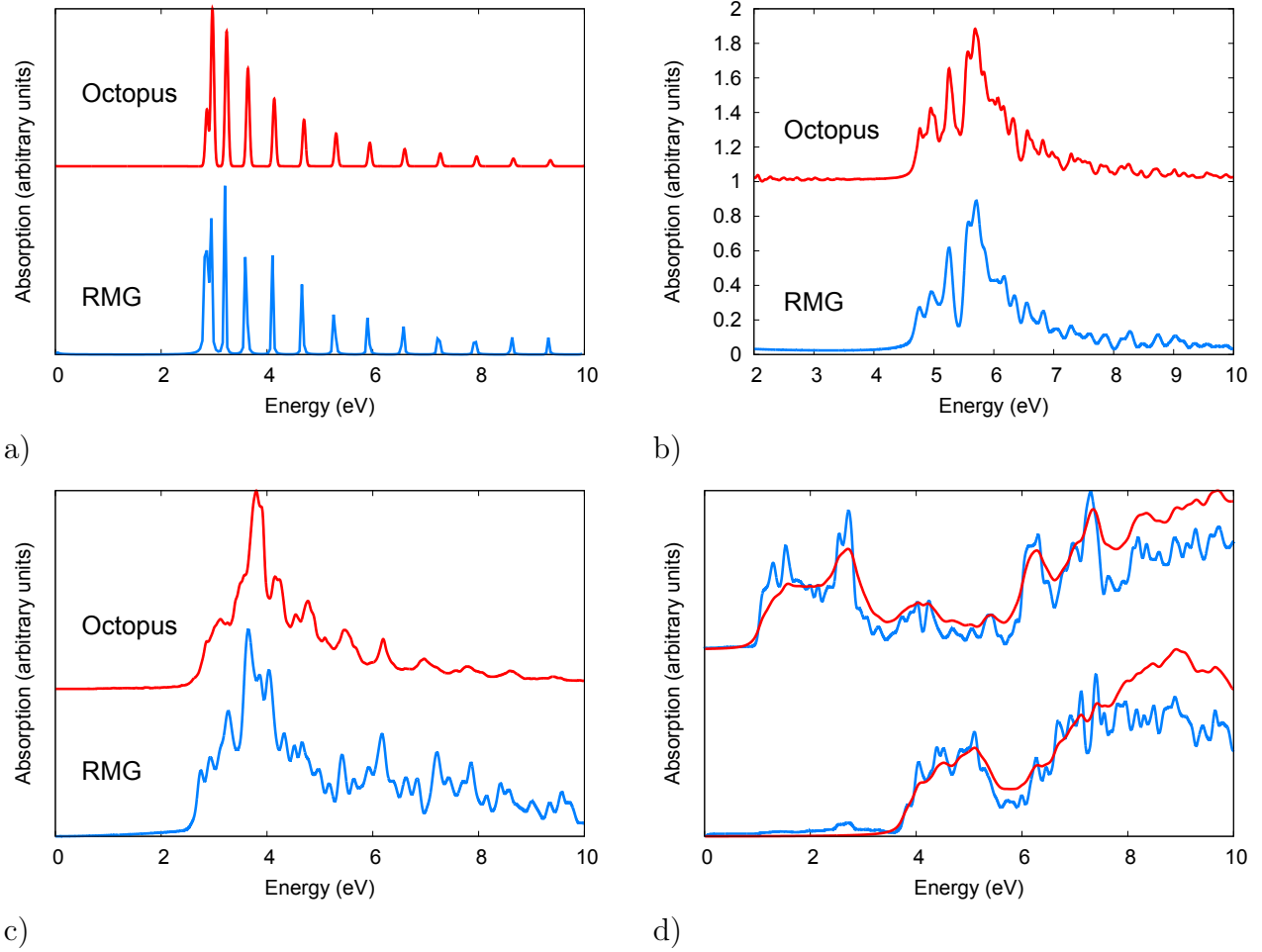


Figure 1: Optical conductivity of (a) 1-D Hydrogen, (b) 2-D hexagonal Boron Nitride (c) 3-D Silicon, (d) spin polarized CrI3 from RMG compared with Octopus

4.2 Two-Dimensional Periodic System: Hexagonal Boron Nitride

Hexagonal boron nitride (hBN) is a wide-gap insulating two-dimensional material with D_{6h} symmetry. Its optical response is highly anisotropic: the in-plane conductivity $\sigma_{xx}(\omega) = \sigma_{yy}(\omega)$ differs markedly from the out-of-plane component $\sigma_{zz}(\omega)$, the latter being negligible for photon energies below the interlayer excitations. This system tests the implementation of two-dimensional periodic boundary conditions and the symmetry enforcement of the conductivity tensor in RMG.

[Computational details: - Monolayer geometry: lattice constant [a] Å, vacuum layer [d] Å to prevent inter-layer interactions - K-point sampling: [N_k × N_k × 1] Monkhorst-Pack - Grid spacing: [value] a_0 - XC functional: [PBE] - Pseudopotential: [type] - Time step and total simulation time: [values]]

[Discussion points: - Agreement between RMG and Octopus for in-plane conductivity - Symmetry: show that $\sigma_{xx} = \sigma_{yy}$ (Fruit 3 from CONTEXT.md) - Physical features: onset at band gap, first excitonic peak if visible - Comparison with available experimental or LR-TDDFT data if relevant]

4.3 Three-Dimensional Periodic System: Silicon

Silicon in the diamond cubic structure ($Fd\bar{3}m$, space group 227) is the canonical benchmark for periodic electronic structure calculations. Its indirect band gap and well-characterized optical spectrum provide a rigorous test of the full three-dimensional periodic RT-TDDFT implementation. The cubic symmetry requires $\sigma_{xx}(\omega) = \sigma_{yy}(\omega) = \sigma_{zz}(\omega)$, providing an immediate symmetry validation.

[Computational details: - Unit cell: conventional 8-atom cell, lattice constant [a] Å - K-point sampling: [N_k × N_k × N_k] Monkhorst-Pack - Grid spacing: [value] a₀ - XC functional: [PBE] - Pseudopotential: [type, NCPP] - Number of occupied + virtual states: [N_occ + N_virt] - Time step Δt : [value] a.u. - Total simulation time: [value] a.u. ([value] fs) - Broadening γ : [value] eV]

[Discussion points: - Agreement between RMG and Octopus: peak positions, lineshapes - Symmetry: $\sigma_{xx} = \sigma_{yy} = \sigma_{zz}$ (Fruit 3) - Energy conservation test for periodic case (Fruit 5 from CONTEXT.md): run longer simulation and show no drift, analogous to paper 1 Fig. 3c - Berry phase comparison if included (Fruit 1) - Physical features: onset near direct gap, main optical peak structure]

4.4 Spin-Polarized System: CrI₃

Chromium triiodide (CrI₃) is a layered van der Waals magnetic insulator with a honeycomb lattice of Cr³⁺ ions, each carrying a local moment of approximately $3\mu_B$. In its ferromagnetic ground state, the spin-up and spin-down electronic structures differ, leading to distinct optical spectra for the two spin channels. This system provides a direct test of the spin-collinear RT-TDDFT extension: the two spin channels should propagate independently but with a common Hartree potential, yielding spin-resolved optical conductivities that can be compared with Octopus and with available experimental data.

[Computational details: - Unit cell: [describe geometry and magnetic ordering] - K-point sampling: [N_k × N_k × N_k] - XC functional: [PBE or PBE+U; specify U if used] - Pseudopotential: [type, NCPP] - Initial spin configuration: ferromagnetic, moments on Cr sites - Time step and total simulation time: [values] - Broadening: [value] eV]

[Discussion points: - Agreement between RMG and Octopus for both spin channels - Physical interpretation: which spectral features are spin-up vs spin-down? Charge-transfer excitations, Cr d-d transitions - Spin current $\mathbf{J}_{\text{spin}} = \mathbf{J}^{\uparrow} - \mathbf{J}^{\downarrow}$ if computed (Fruit 4 from CONTEXT.md) - Comparison with available experimental optical data if relevant - Note: ECD/MCD spectra not computed in this work (magnetic perturbation formalism reserved for future publication)]

4.5 JACEK: Large-Scale Demonstration: NV Center in Diamond

As a final demonstration, we apply the periodic spin-polarized RT-TDDFT implementation to the negatively charged nitrogen-vacancy (NV⁻) center in diamond, one of the most studied point defects in solid-state physics.^{???} The ground state of NV⁻ center is an optically addressable spin-triplet located within the diamond band gap^{???} Because of its optical addressability and long coherence times at room temperature the NV(-) center is an attrac-

tive system for several emerging quantum technologies, including single-photon emitter, nanoscale magnetometry, quantum sensing, quantum information processing. ? ? ? ? ?

This system tests several capabilities simultaneously: spin-polarized real-time propagation, optical response of a localized defect in a three-dimensional periodic host, convergence toward the dilute-defect limit, as well as computational scaling to thousands of atoms. The purpose of this section is not to provide a complete quantitative description of NV-photophysics. Instead, the present calculations examine the electronic optical response of defect in the energy region of the experimental zero-phonon line (ZPL) and use the NV⁻ center to demonstrate how polarization selection rules, periodic defect-image interactions, and active-space truncation affect large-scale periodic RT-TDDFT calculations.

4.5.1 Electronic structure and simulation setup for NV⁻

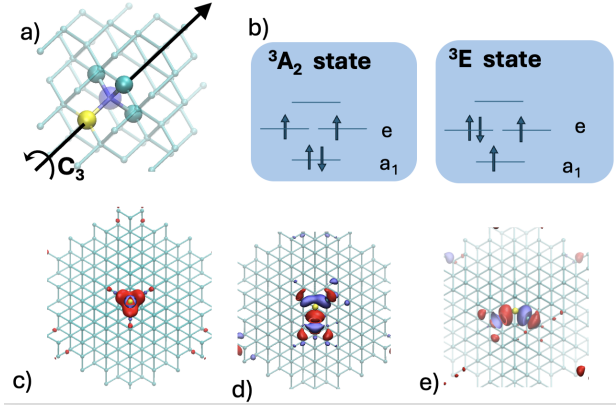


Figure 2: Electronic structure of the negatively charged nitrogen-vacancy (NV⁻) center in diamond. (a) Local atomic structure of the defect, consisting of a substitutional nitrogen atom (green) adjacent to a carbon vacancy (blue transparent). The black arrow indicates the [111] direction, which defines the C_3 symmetry axis of the approximately C_{3v} defect environment. (b) Schematic occupation of the defect levels in the triplet ground state, 3A_2 (configuration $a_1^2 e_1^1$), and in the optically excited triplet 3E state (configuration $a_1^1 e_2^2$). The defect-associated optical transition in the experimental zero-phonon-line energy region is dominated by promotion of a minority-spin electron from the occupied a_1 -like defect orbital into the e manifold. (c-e) Representative real-space isosurfaces of Kohn-Sham defect orbitals localized in the diamond band gap (view along the C_3 axis): (c) the occupied a_1 -like orbital and (d,e) the two components of the nearly degenerate e manifold. Red and blue isosurfaces denote opposite phases of the Kohn-Sham orbitals.

The NV⁻ center consists of a substitutional nitrogen atom adjacent to a carbon vacancy, with one additional electron trapped at the defect (Figure 2a). The defect axis is oriented along the crystallographic [111] direction, which defines the C_3 symmetry axis of the approximately C_{3v} local defect environment. The defect states are located within the diamond band gap. The lower a_1 -like defect orbital has substantial nitrogen character together with contributions from the three carbon atoms adjacent to the vacancy, whereas the nearly degenerate e manifold is primarily localized on these three carbon atoms, as shown in Figure 2c-e. Because pristine diamond has a wide band gap of about 5.5 eV and is transparent throughout the visible range, optical absorption and emission below this energy arise from defect

states; for NV^- , the ${}^3A_2 \rightarrow {}^3E$ transition has an experimental zero-phonon line at 1.945 eV, corresponding to 637 nm red light.

All NV^- calculations were performed here with the **RMG** code using periodic boundary conditions, the PBE exchange-correlation functional, and the **NC_accuracy** norm-conserving pseudopotentials. The diamond lattice constant was first optimized for bulk diamond using the same functional and pseudopotentials, giving $a = 3.5725125$ Å. This optimized lattice constant was then kept fixed for all NV^- defect calculations.

Three single-defect supercell sizes were considered, denoted $\text{NV-}x2$ (small), $\text{NV-}x4$ (medium), and $\text{NV-}x8$ (large). These systems were constructed from $2 \times 2 \times 2$, $4 \times 4 \times 4$, and $8 \times 8 \times 8$ repetitions of the conventional face-centered cubic (fcc) diamond cell, corresponding to 64, 512, and 4096 diamond lattice sites before formation of the defect. Previous supercell studies of NV^- in diamond have shown that relatively large cells, on the order of the 512-site $\text{NV-}x4$ model, are needed to reduce spurious interactions between periodic defect images.^{???} Here we directly test this finite-size behavior by extending the real-time calculations to the 4096-site $\text{NV-}x8$ cell. The same 4096-site host also enables explicit simulations of periodic arrays of NV^- centers, including collinear parallel and antiparallel defect-spin arrangements, which are used below as an additional demonstration of spin-resolved periodic RT-TDDFT. The actual defect supercells contained 63, 511, and 4095 atoms, respectively. The negatively charged supercells were treated using a uniform compensating background charge to maintain overall charge neutrality under periodic boundary conditions.

The real-space grid was scaled with the supercell size so that all three calculations used the same grid spacing. The $\text{NV-}x2$, $\text{NV-}x4$, and $\text{NV-}x8$ systems used 48^3 , 96^3 , and 192^3 grid points, respectively, corresponding to a grid spacing of $0.281 a_0$. This grid corresponds to equivalent wavefunction and charge-density cutoffs of 124.7 and 498.9 Ry, respectively; the charge-density grid is two times finer in each direction.

After construction of the charged spin-polarized defect cells, all atomic coordinates were relaxed at fixed lattice vectors. For each supercell size, the spin-polarized triplet solution was found to be lower in energy than the corresponding singlet solution, and the relaxed triplet geometries were used as the starting point for the RT-TDDFT simulations. All real-time calculations used spin-collinear density-matrix propagation with a time step of 1.0 atomic unit of time and production trajectories of 40000 time steps, corresponding to approximately 1.0 ps.

Table 2: Single-defect NV^- supercells used in the RT-TDDFT calculations. The number of diamond sites refers to the pristine diamond supercell before creating the NV center. The actual number of atoms is one smaller after replacing one C atom by N and removing a nearest-neighbor C atom to form the vacancy.

System	Supercell	Sites	Atoms	Grid	$h (a_0)$	k -point sampling
$\text{NV-}x2$ (small)	$2 \times 2 \times 2$	64	63	48^3	0.281	Γ , $4 \times 4 \times 4$
$\text{NV-}x4$ (medium)	$4 \times 4 \times 4$	512	511	96^3	0.281	Γ , $2 \times 2 \times 2$
$\text{NV-}x8$ (large)	$8 \times 8 \times 8$	4096	4095	192^3	0.281	Γ

4.5.2 Polarization selection rules and spin selectivity

The polarization-resolved RT-TDDFT spectra provide a direct test of the local C_{3v} selection rules of the NV^- center. This analysis was performed for the medium-size single-defect $NV-x4$ supercell (511 atoms) at the Γ point. The TDDFT active space consisted of 1024 occupied and 3200 virtual orbitals (approximately 6 per each C atom), overall total 4324 orbitals. Three impulsive electric-field polarizations were considered: one parallel to the NV axis,

$$\hat{\mathbf{e}}_{\parallel} = \frac{1}{\sqrt{3}}(1, 1, 1), \quad (13)$$

and two mutually orthogonal directions perpendicular to the NV axis,

$$\hat{\mathbf{e}}_{\perp,1} = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \hat{\mathbf{e}}_{\perp,2} = \frac{1}{\sqrt{6}}(1, 1, -2). \quad (14)$$

The calculated spectra show that the defect-associated peak in the experimental zero-phonon-line energy region is absent for the polarization parallel to the $[111]$ defect axis, while it appears clearly for both transverse polarizations, $\hat{\mathbf{e}}_{\perp,1}$ and $\hat{\mathbf{e}}_{\perp,2}$. See Figure 3.

This behavior follows directly from the symmetry of the ${}^3A_2 \rightarrow {}^3E$ transition. In the local C_{3v} point group, the velocity (or dipole operator for length gauge case) component parallel to the defect axis transforms as A_1 , whereas the two transverse velocity (or dipole operator for length gauge case) components transform together as the two-dimensional E representation. The electric-dipole matrix element is symmetry allowed only when the direct product of the initial state, dipole operator, and final state contains the totally symmetric representation A_1 . For axial polarization,

$$A_2 \otimes A_1 \otimes E = E, \quad (15)$$

which does not contain A_1 , and the transition is forbidden. For transverse polarization,

$$A_2 \otimes E \otimes E = A_1 \oplus A_2 \oplus E, \quad (16)$$

which contains A_1 , and the transition is allowed. The computed polarization dependence is therefore consistent with the expected selection rule that the optical transition dipole for the NV^- ${}^3A_2 \rightarrow {}^3E$ transition lies perpendicular to the NV axis.

The same spectra also demonstrate spin selectivity of the defect-region optical response. In the collinear spin convention used in the present calculations, the peak in the zero-phonon-line energy region appears predominantly in the spin-down channel. Equivalently, this is the minority-spin channel of the triplet configuration shown in Fig. 2b. The dominant excitation promotes a minority-spin electron from the occupied a_1 -like defect orbital into the e manifold, producing the configuration schematically shown for the optically excited triplet state. Because the electric-dipole perturbation does not flip spin, the transition appears in the spin channel associated with this minority-spin $a_1 \rightarrow e$ excitation. This spin-resolved selection rule provides a stringent test of the spin-collinear RT-TDDFT implementation for localized defect states in a periodic host.

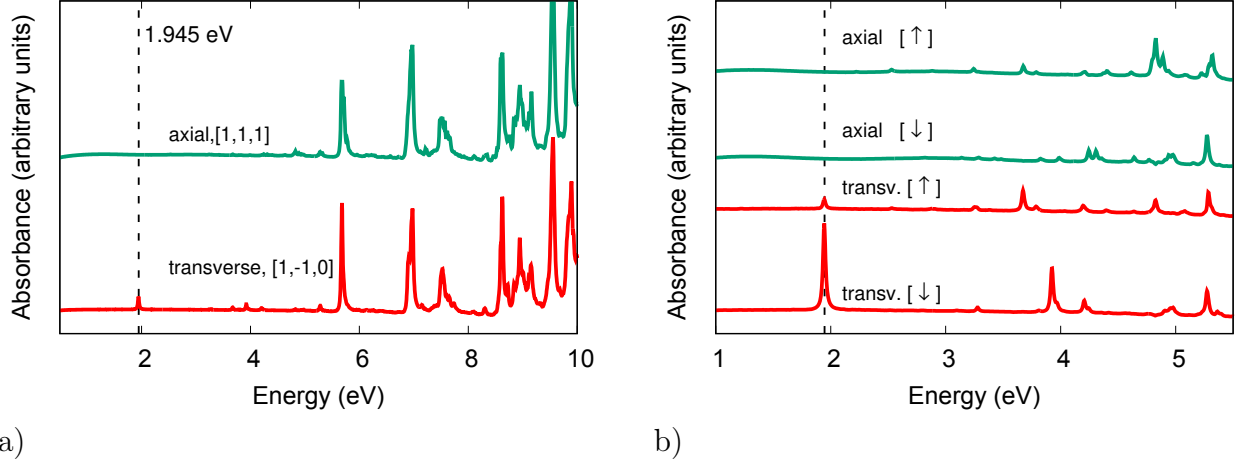


Figure 3: Polarization and spin selectivity of the NV^- defect optical response from spin-collinear RT-TDDFT for the 512-site $\text{NV-}x4$ supercell. (a) Spin-summed spectra for axial and transverse light polarizations. The axial spectrum corresponds to a kick polarized parallel to the NV axis, $\mathbf{E} \parallel [111]$, while the transverse spectrum corresponds to $\mathbf{E} \parallel [1\bar{1}0]$, perpendicular to the NV axis. The dashed vertical line marks the experimental NV^- zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The defect-associated peak near this energy is strongly enhanced for transverse polarization and suppressed for axial polarization, consistent with the C_{3v} selection rule for the ${}^3A_2 \rightarrow {}^3E$ transition. (b) Expanded view of the sub-gap optical response below approximately 5.5 eV, corresponding to the transparency window of pristine diamond. In this energy range, optical features arise from the NV^- defect states rather than from intrinsic band-to-band absorption of the diamond host. The spin-resolved spectra show that the transverse response near the ZPL energy is dominated by the spin-down, minority-spin channel, consistent with the dominant $a_1 \rightarrow e$ defect excitation shown schematically in Fig. 2. The spectra are vertically shifted for clarity.

4.5.3 Active-space convergence of the defect excitation

The selection-rule analysis above used the medium-size single-defect $\text{NV-}x4$ supercell at the Γ point with the TDDFT active space including full occupied orbitals manifold and 3200 virtual orbitals. Here we use the same system and transverse polarization to examine how the defect-associated peak depends on the active propagation manifold used in the RT-TDDFT calculation. As the system size increases, the number of occupied and unoccupied Kohn–Sham states included in the density-matrix propagation can grow rapidly, making the computational cost prohibitive. Similarly, one might expect the low-energy defect excitation to be only weakly sensitive to deep occupied states or to high-energy virtual states, since the relevant NV^- defect states are localized in the diamond band gap. We investigate the effect of the TDDFT active space construction on position of the experimental zero phonon line 1.945 eV peak to quantify the effects of this effect numerically in terms of fraction of the occupied orbitals, $f_{\text{occ}} = \frac{N_{\text{occ}}^{\text{act}}}{N_{\text{occ}}^{\text{full}}}$, and the number of virtual states per diamond site, $n_{\text{virt}} = \frac{N_{\text{virt}}}{N_{\text{site}}}$, used to construct the TDDFT active space. Here, $N_{\text{occ}}^{\text{full}}$ represents the total number of occupied orbitals, $N_{\text{occ}}^{\text{act}}$ is the number of occupied states retained in the propagation, N_{virt} is the number of unoccupied states included in the active manifold and N_{site} is the total number of diamond lattice sites before formation of the defect.

For the 512-site $\text{NV-}x4$ supercell used in this convergence test, $N_{\text{occ}}^{\text{full}} = 1024$ and $N_{\text{site}} = 512$. The occupied-space tests removed the lowest 0, 200, 400, 512, 600, and 800 occupied

Kohn–Sham states from the active propagation manifold. This corresponds to retaining 1024, 824, 624, 512, 424, and 224 occupied states, or approximately 100%, 80%, 60%, 50%, 40%, and 20% of the occupied manifold, respectively. The virtual-space tests used $N_{\text{virt}} = 512, 1024, 2048$, and 3200, corresponding to $n_{\text{virt}} = 1, 2, 4$, and 6.25 virtual states per diamond site.

Figure 4 shows the effect of occupied- and virtual-space truncation on the position of the defect-associated peak near the experimental ZPL energy. Reducing f_{occ} shifts the peak to higher energy, showing that the deep occupied manifold cannot be discarded without changing the computed defect excitation. Conversely, increasing n_{virt} systematically shifts the peak to lower energy, with the peak position approaching a saturated value as the virtual space is enlarged. Physically, this behavior indicates that the localized defect excitation is dressed by the polarizable diamond environment and by the self-consistent time-dependent response of the electron density. Truncating either the occupied or unoccupied part of the propagation manifold suppresses electronic relaxation and screening around the particle-hole excitation, producing a measurable blueshift

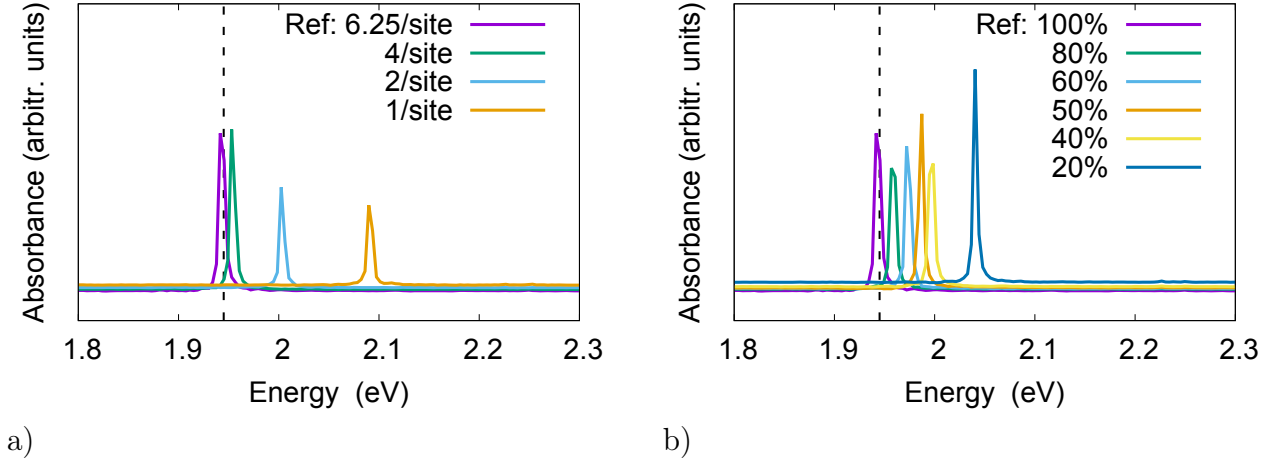


Figure 4: Active-space convergence of the defect-associated optical response for the 512-site NV- $x4$ supercell at the Γ point, using transverse polarization along $[1\bar{1}0]$ direction. The dashed vertical line marks the experimental NV⁻ zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The reference calculation, shown in both panels, uses the full occupied space, $f_{\text{occ}} = 100\%$, and $N_{\text{virt}} = 3200$, corresponding to $n_{\text{virt}} = 6.25$ virtual states per diamond site for the 512-site cell. (a) Dependence on the virtual active space, expressed as the number of virtual states per diamond site, $n_{\text{virt}} = N_{\text{virt}}/N_{\text{site}}$. The values $n_{\text{virt}} = 1, 2, 4$, and 6.25 correspond to $N_{\text{virt}} = 512, 1024, 2048$, and 3200, respectively. Increasing n_{virt} systematically red-shifts the peak toward a saturated value. (b) Dependence on the retained occupied fraction, $f_{\text{occ}} = N_{\text{occ}}^{\text{act}}/N_{\text{occ}}^{\text{full}}$. The retained values 100%, 80%, 60%, 50%, 40%, and 20% are approximate and correspond to 1024, 800, 600, 512, 400, and 200 occupied states. Reducing f_{occ} shifts the defect-associated peak to higher energy.

Furthermore, the two convergence tests show different sensitivities. At fixed $n_{\text{virt}} = 6.25$, reducing the retained occupied fraction from 100% to about 50% produces only a modest blue shift of the defect-associated peak, while retaining only about 20% of the occupied space leads to a larger shift. By contrast, at fixed $f_{\text{occ}} = 100\%$, increasing the virtual active space from $n_{\text{virt}} = 1$ to 4–6.25 virtual states per site produces a substantially larger red shift and brings the peak close to the experimental ZPL energy. Thus, for this NV- $x4$ test case, the

unoccupied portion of the active space is the primary convergence bottleneck for the peak position, while the occupied manifold can be moderately truncated with a smaller, but still measurable, effect.

In summary, the peak position is more sensitive to the size of the virtual active space than to moderate truncation of the occupied manifold. However, aggressive occupied-space truncation also produces a measurable blue shift, indicating that the response of deeper valence states contributes to the screening and relaxation of the localized defect excitation.

4.5.4 Defect concentration and magnetic alignment in replicated NV⁻ arrays

The large NV-*x*8 supercell also provides a controlled framework for examining how the optical response evolves with defect concentration. Starting from the same 4096-site diamond host, we constructed cells containing 1, 8, and 64 NV⁻ centers. These systems correspond, respectively, to the defect concentrations obtained from the NV-*x*8, NV-*x*4, and NV-*x*2 cells: the 8-NV structure is generated by a $2 \times 2 \times 2$ replication of the NV-*x*4 cell, whereas the 64-NV structure is generated by a $4 \times 4 \times 4$ replication of the NV-*x*2 cell. The comparison therefore changes the number and separation of the defects while retaining the same total supercell size, real-space grid, transverse field polarization, spectral broadening, and Γ -point representation. The concentration series discussed first uses a parallel collinear alignment of the NV moments.

The RT-TDDFT calculations for the three 4096-site systems also used the same active-space density. Approximately 67% of the occupied space was retained, while the virtual space contained approximately two unoccupied states per diamond site. In the RMG state-count convention, the calculations used 16400 computed states, with 2700 deep occupied states frozen and four highest-energy states discarded, leaving 13696 states in the density-matrix propagation. Each spectrum was obtained from 42000 electronic time steps with $\Delta t = 1.0$ a.u., corresponding to a total propagation time of approximately 1.02 ps. Consequently, differences among the three RT-TDDFT spectra predominantly reflect the NV concentration rather than changes in total cell size, propagation length, or active-space density.

For comparison, we also evaluated the optical response within the independent-particle approximation (IPA) using the LOPTICS implementation in RMG. In atomic units, the spin-resolved imaginary part of the IPA dielectric tensor can be written as

$$\epsilon_{\alpha\beta}^{(2),\sigma}(\omega) = \frac{4\pi^2}{\Omega\omega^2} \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{v,c} (f_{v\mathbf{k}\sigma} - f_{c\mathbf{k}\sigma}) \text{Re} \left[P_{v\mathbf{k}\sigma}^{\alpha*} P_{c\mathbf{k}\sigma}^{\beta} \right] \delta_{\eta} (\epsilon_{c\mathbf{k}\sigma} - \epsilon_{v\mathbf{k}\sigma} - \omega), \quad (17)$$

where Ω is the supercell volume, $w_{\mathbf{k}}$ is the $\mathbf{k}$ -point weight, and $P_{v\mathbf{k}\sigma}^{\alpha}$ is the velocity-gauge transition matrix element, including the nonlocal-pseudopotential contribution. The indices v and c denote occupied and unoccupied Kohn–Sham states of the same global spin σ , and δ_{η} represents the broadened energy-conservation condition. The IPA spectra therefore expose the underlying vertical Kohn–Sham transition structure, whereas the RT-TDDFT spectra additionally contain the self-consistent induced Hartree and adiabatic exchange–correlation response. The IPA curves in Fig. 5 were independently scaled for clarity, so the comparison concerns peak positions and spectral structure rather than absolute intensities.

Figure 5 shows a pronounced evolution of the optical response with decreasing NV con-

centration. For the 64-NV array, corresponding to the NV- $x2$ concentration, both IPA and RT-TDDFT produce strongly fragmented spectra over the sub-gap energy range. The large number of optically active features is consistent with substantial coupling among the closely spaced defects, which hybridizes the defect-derived states and produces dispersive occupied and unoccupied defect bands. The NV- $x2$ concentration is therefore clearly outside the isolated-defect regime.

At the intermediate NV- $x4$ concentration, represented by eight defects in the same 4096-site host, the spectral response becomes much simpler. The IPA spectrum retains several closely spaced transitions in the vicinity of the experimental ZPL energy, indicating that some fine structure remains in the underlying Kohn–Sham transition spectrum. By contrast, the self-consistent RT-TDDFT response is dominated by a single resonance near 2.01 eV. This simplification is consistent with coupling and redistribution of oscillator strength among nearby independent-particle transitions in the time-dependent response. Finite spectral resolution and the truncated RT-TDDFT active space may also suppress weaker components, so the single RT-TDDFT peak should not be interpreted as evidence that every periodic-image effect has disappeared. Nevertheless, it shows that the dominant optical observable is already substantially single-resonance-like at the NV- $x4$ defect separation.

Further dilution to a single NV⁻ center in the 4096-site NV- $x8$ cell produces a similarly simple low-energy response. The qualitative similarity of the dominant RT-TDDFT peaks for the 8-NV and 1-NV systems supports the conclusion that a separation corresponding to the 512-site NV- $x4$ model is sufficient to recover the principal defect-associated optical feature within the present spectral resolution and active-space approximation. The calculated peak near 2.01 eV remains approximately 0.06 eV above the experimental 1.945-eV ZPL. As shown by the active-space analysis in Sec. 5.5.4, part of this residual blueshift is consistent with the restricted occupied and virtual propagation spaces used for the 4096-site calculations. The experimental ZPL is nevertheless used only as a reference because the present frozen-ion adiabatic-PBE response also omits excited-state structural relaxation and vibronic effects.

The fixed-volume replicated cells also permit examination of how the magnetic arrangement of the NV moments affects the spin-resolved optical response. We considered both a parallel collinear configuration, denoted FM below, and a G-type antiparallel broken-symmetry configuration, denoted G-AFM, for the 64-NV and 8-NV arrays. In the FM configuration, all local defect moments are aligned with the same global spin axis. In the G-AFM configuration, the local moment orientation alternates between neighboring defect sites.

The persistence of the local moments in these configurations can be characterized using the total and absolute magnetizations,

$$M_{\text{tot}} = \mu_B \int [\rho_{\uparrow}(\mathbf{r}) - \rho_{\downarrow}(\mathbf{r})] d\mathbf{r}, \quad M_{\text{abs}} = \mu_B \int |\rho_{\uparrow}(\mathbf{r}) - \rho_{\downarrow}(\mathbf{r})| d\mathbf{r}. \quad (18)$$

For the 8-NV array, the FM solution gives $M_{\text{tot}} = 20.73 \mu_B$ and $M_{\text{abs}} = 22.36 \mu_B$, whereas the G-AFM solution gives $M_{\text{tot}} = 0$ and $M_{\text{abs}} = 18.66 \mu_B$. The corresponding values for the 64-NV array are $M_{\text{tot}} = 128.42 \mu_B$ and $M_{\text{abs}} = 143.03 \mu_B$ for the FM arrangement, and $M_{\text{tot}} = 0$ and $M_{\text{abs}} = 138.57 \mu_B$ for the G-AFM arrangement. The vanishing total magnetization but substantial absolute magnetization of the G-AFM solutions confirms that

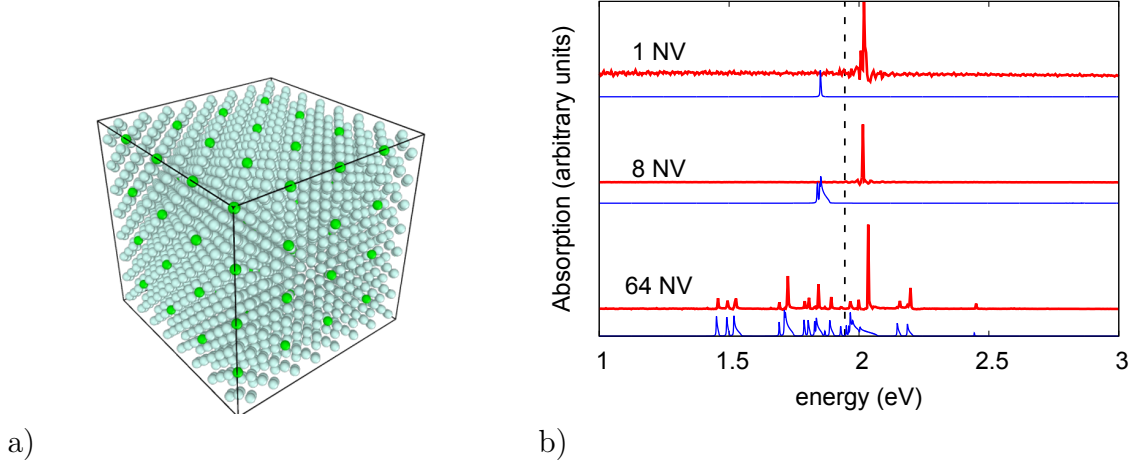


Figure 5: Concentration dependence of the optical response in parallel-aligned NV^- arrays. Spin-collinear RT-TDDFT spectra are shown in red and independent-particle approximation (IPA) spectra, evaluated with the LOPTICS implementation in RMG, are shown in blue. From top to bottom, the systems contain 1, 8, and 64 NV^- centers in the same 4096-site diamond host, corresponding to the NV- x 8, NV- x 4, and NV- x 2 defect concentrations, respectively. All calculations use the same atomic structures, real-space grid, Γ -point representation, transverse polarization, and spectral broadening. The RT-TDDFT calculations additionally use the same retained occupied fraction and number of virtual states per diamond site and were propagated for 42000 steps with $\Delta t = 1.0$ a.u., corresponding to approximately 1.02 ps. The dashed vertical line marks the experimental zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The strongly fragmented spectrum of the 64-NV array reflects substantial coupling among closely spaced defects. At the 8-NV concentration, the IPA retains nearby independent-particle transition structure, whereas the RT-TDDFT response is dominated by a single resonance resembling that of the dilute single-NV system. The spectra are vertically offset, and the IPA curves are independently scaled, for clarity.

the local defect moments remain present and cancel globally rather than disappearing.

The spin-resolved spectra in Fig. 6 show how the local minority-spin selection rule established for the single NV center maps onto the globally defined spin channels of a multi-defect array. In the FM configuration, all local moments have the same orientation. The local minority-spin $a_1 \rightarrow e$ excitation therefore corresponds to the same global spin channel at every defect, and the optical response is strongly dominated by one spin component. In the G-AFM arrangement, the local moment is reversed on the second magnetic sublattice. The local minority-spin excitation consequently maps onto one global spin channel on one sublattice and the opposite global spin channel on the other. The corresponding defect-associated features therefore appear in both global spin-resolved spectra.

This redistribution between global spin channels does not represent a spin-flip optical transition. The optical perturbation remains spin independent and the transitions conserve the global collinear spin label. Instead, the result reflects the reversal of the local moment relative to the fixed global quantization axis. Local minority spin is represented by opposite global spin labels on the two G-AFM sublattices.

The concentration dependence found above is also evident in the magnetic comparison. At the NV- x 2 concentration, both FM and G-AFM spectra contain many defect-associated peaks, and the detailed distribution of these features depends on the magnetic arrangement. At the more dilute NV- x 4 concentration, the response is concentrated primarily in a single

resonance near 2.0 eV. Changing from FM to G-AFM then affects mainly the distribution of this resonance between the two global spin channels rather than producing a comparably strong fragmentation of the spectrum. This behavior is consistent with a substantial reduction of electronic coupling among the NV centers as their separation increases.

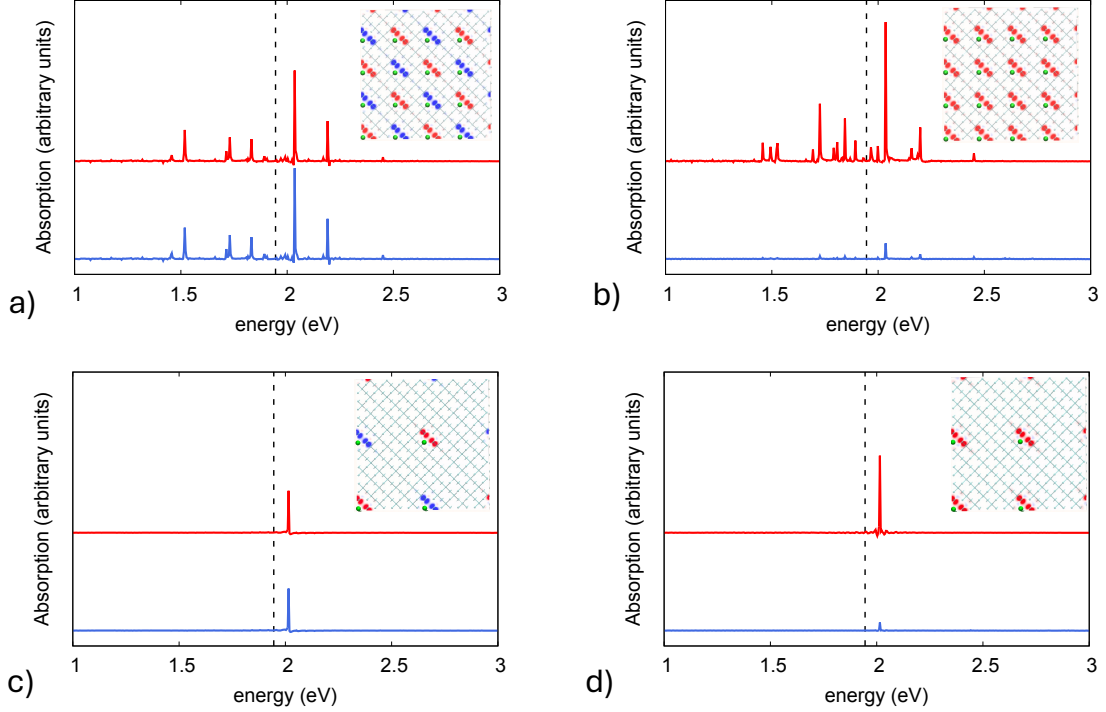


Figure 6: Spin-resolved RT-TDDFT optical response of replicated NV^- arrays in the 4096-site diamond host. The upper row shows the 64-NV system at the $\text{NV-}x2$ concentration, and the lower row shows the 8-NV system at the $\text{NV-}x4$ concentration. Panels (a) and (c) correspond to G-type antiparallel broken-symmetry arrangements, whereas panels (b) and (d) correspond to parallel, FM arrangements. The red and blue curves show the two globally defined collinear spin channels and are vertically offset for clarity. The insets illustrate the corresponding local defect-spin arrangements; red and blue markers denote opposite local moment orientations. The dashed vertical lines mark the experimental NV^- zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. In the FM configurations, the local minority-spin defect excitation maps predominantly onto one global spin channel. In the G-AFM configurations, reversal of the local moments between the two magnetic sublattices causes the same local minority-spin excitation to appear in both global spin channels. The response remains strongly fragmented at the $\text{NV-}x2$ concentration but becomes predominantly single-resonance-like at the $\text{NV-}x4$ concentration.

Together, these calculations show that increasing the NV–NV separation transforms the optical response from the fragmented spectrum of an interacting defect array to a dominant isolated-defect-like resonance. At the same time, the magnetic alignment determines how the local minority-spin excitation is represented in the globally defined spin channels. The 4096-site calculations therefore demonstrate that the periodic spin-collinear RT-TDDFT implementation can treat both dilute defects and extended arrays containing as many as 64 optically active defect centers within the same real-space simulation framework.

5 Numerical Results

We benchmark the periodic and spin-polarized RT-TDDFT implementation in **RMG** against the **Octopus** code[?] for four representative systems of increasing dimensionality and physical complexity: a one-dimensional hydrogen chain (H_2), a two-dimensional hexagonal boron nitride (hBN) monolayer, a three-dimensional silicon crystal, and the spin-polarized magnetic insulator CrI_3 . These systems are chosen to isolate and validate specific aspects of the implementation: k-point sampling, anisotropic optical response, three-dimensional periodic boundaries, and spin-collinear propagation, respectively. The section concludes with a demonstration of the unique large-scale capability of **RMG** through calculations on nitrogen-vacancy (NV) centers in diamond at varying supercell sizes.

In all calculations, a momentum kick of amplitude $E_0 = [\text{value}]$ a.u. was applied at $t = 0$ along each Cartesian direction independently, and the optical conductivity was obtained from the Fourier transform of the resulting current density. [Add: common computational parameters – time step, total propagation time, broadening gamma, XC functional, pseudopotential library used for all systems.]

5.1 One-Dimensional Periodic System: Hydrogen Chain

As the simplest possible periodic test system, we consider an infinite chain of hydrogen molecules aligned along the z axis with alternating bond lengths, modeled by a one-dimensional supercell with periodic boundary conditions. This system provides a stringent test of k-point sampling and the velocity gauge implementation in the limit of a quasi-one-dimensional geometry, where the longitudinal (z -direction) and transverse (x, y -direction) optical responses differ substantially.

[Computational details: - Supercell geometry: H–H bond length, H_2 – H_2 separation - K-point sampling: $[\text{N_k}]$ points along z , Gamma in x and y - Grid spacing: $[\text{value}]$ a_0 - XC functional: [PBE or LDA] - Pseudopotential: $[\text{type}]$ - Time step Δt : $[\text{value}]$ a.u. - Total simulation time: $[\text{value}]$ a.u. - Broadening γ : $[\text{value}]$ eV]

[Discussion points: - Agreement between **RMG** and **Octopus** for both longitudinal and transverse components - K-point convergence: how many k-points needed for converged spectrum? - Physical interpretation of the spectral features - If Berry phase comparison is available: show agreement between current-integration and BP polarization as internal consistency check (Fruit 1 from CONTEXT.md) - If ground-state current convergence is shown: include as panel or note in text (Fruit 2 from CONTEXT.md)]

5.2 Two-Dimensional Periodic System: Hexagonal Boron Nitride

Hexagonal boron nitride (hBN) is a wide-gap insulating two-dimensional material with D_{6h} symmetry. Its optical response is highly anisotropic: the in-plane conductivity $\sigma_{xx}(\omega) = \sigma_{yy}(\omega)$ differs markedly from the out-of-plane component $\sigma_{zz}(\omega)$, the latter being negligible for photon energies below the interlayer excitations. This system tests the implementation of two-dimensional periodic boundary conditions and the symmetry enforcement of the conductivity tensor in **RMG**.

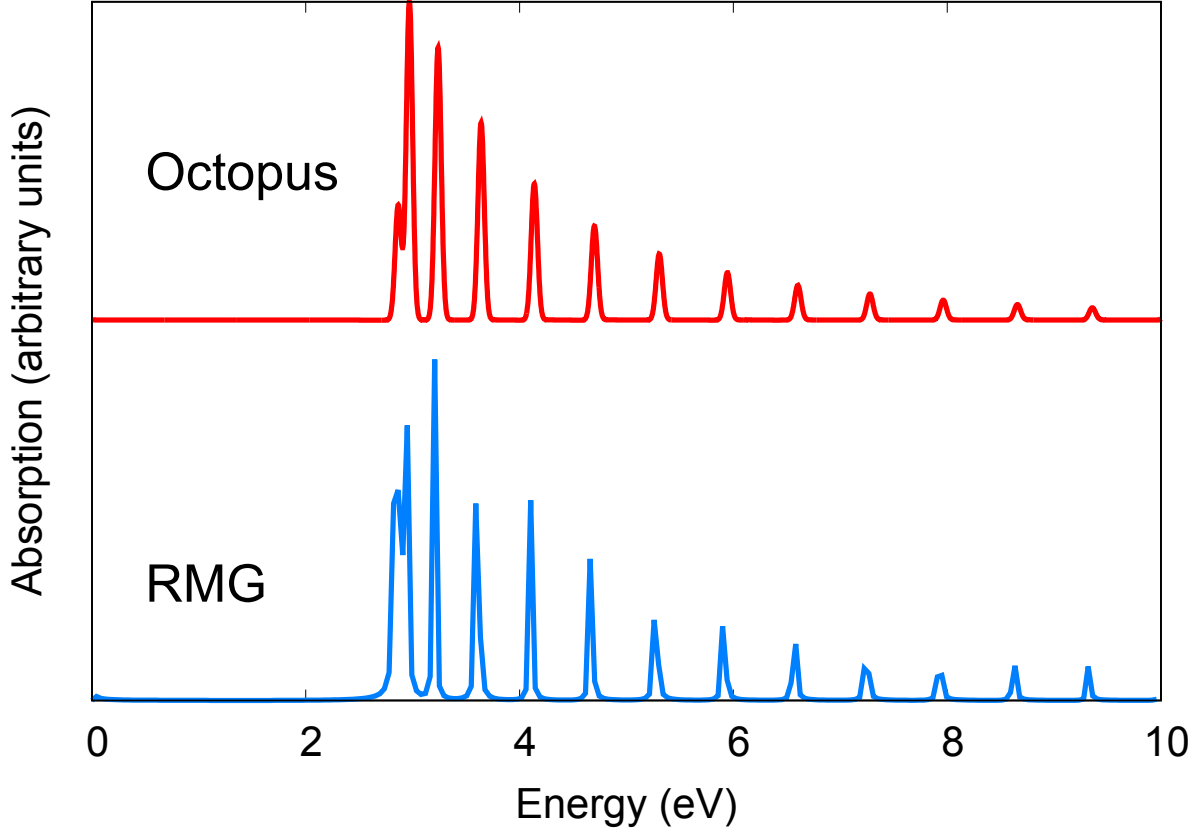


Figure 7: [Caption for H₂ chain optical conductivity figure. Should describe: (a) longitudinal $\sigma_{zz}(\omega)$ and transverse $\sigma_{xx}(\omega)$ components from RMG compared with Octopus; (b) convergence with k-point density if shown; labels for RMG and Octopus curves. Note anisotropy of 1D system as key feature to highlight.]

[Computational details: - Monolayer geometry: lattice constant [a] Å, vacuum layer [d] Å to prevent inter-layer interactions - K-point sampling: [N_k × N_k × 1] Monkhorst-Pack - Grid spacing: [value] a₀ - XC functional: [PBE] - Pseudopotential: [type] - Time step and total simulation time: [values]]

[Discussion points: - Agreement between RMG and Octopus for in-plane conductivity - Symmetry: show that $\sigma_{xx} = \sigma_{yy}$ (Fruit 3 from CONTEXT.md) - Physical features: onset at band gap, first excitonic peak if visible - Comparison with available experimental or LR-TDDFT data if relevant]

5.3 Three-Dimensional Periodic System: Silicon

Silicon in the diamond cubic structure ($Fd\bar{3}m$, space group 227) is the canonical benchmark for periodic electronic structure calculations. Its indirect band gap and well-characterized optical spectrum provide a rigorous test of the full three-dimensional periodic RT-TDDFT implementation. The cubic symmetry requires $\sigma_{xx}(\omega) = \sigma_{yy}(\omega) = \sigma_{zz}(\omega)$, providing an immediate symmetry validation.

[Computational details: - Unit cell: conventional 8-atom cell, lattice constant [a] Å- K-

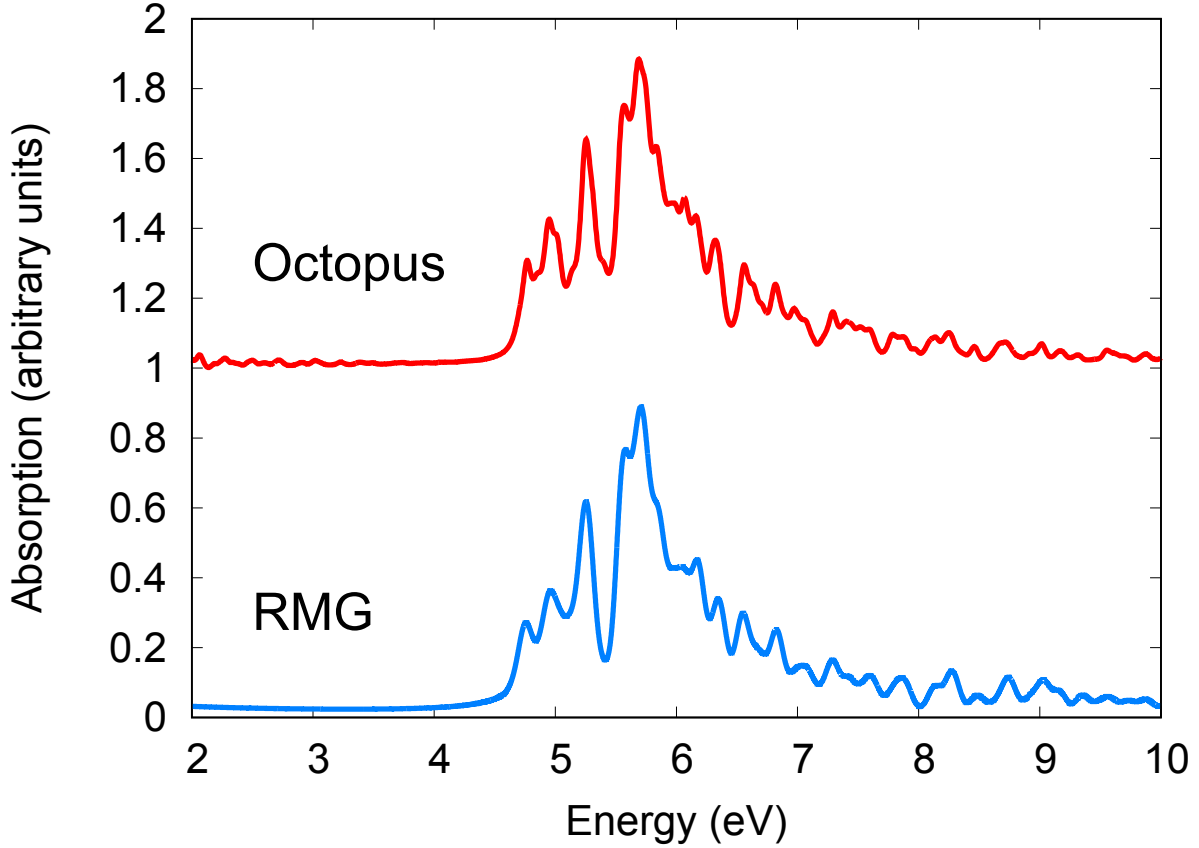


Figure 8: [Caption for hBN optical conductivity figure. Should describe: in-plane $\sigma_{xx}(\omega)$ from RMG and Octopus; out-of-plane component if shown; indicate the onset of absorption near the band gap. Highlight that $\sigma_{xx} = \sigma_{yy}$ (symmetry correctly reproduced) and $\sigma_{zz} \approx 0$.]

point sampling: $[N_k \times N_k \times N_k]$ Monkhorst-Pack - Grid spacing: [value] a_0 - XC functional: [PBE] - Pseudopotential: [type, NCPP] - Number of occupied + virtual states: $[N_{occ} + N_{virt}]$ - Time step Δt : [value] a.u. - Total simulation time: [value] a.u. ([value] fs) - Broadening γ : [value] eV]

[Discussion points: - Agreement between RMG and Octopus: peak positions, lineshapes
- Symmetry: $\sigma_{xx} = \sigma_{yy} = \sigma_{zz}$ (Fruit 3) - Energy conservation test for periodic case (Fruit 5 from CONTEXT.md): run longer simulation and show no drift, analogous to paper 1 Fig. 3c
- Berry phase comparison if included (Fruit 1) - Physical features: onset near direct gap, main optical peak structure]

5.4 Spin-Polarized System: CrI_3

Chromium triiodide (CrI_3) is a layered van der Waals magnetic insulator with a honeycomb lattice of Cr^{3+} ions, each carrying a local moment of approximately $3\mu_B$. In its ferromagnetic ground state, the spin-up and spin-down electronic structures differ, leading to distinct optical spectra for the two spin channels. This system provides a direct test of the spin-collinear RT-TDDFT extension: the two spin channels should propagate independently but

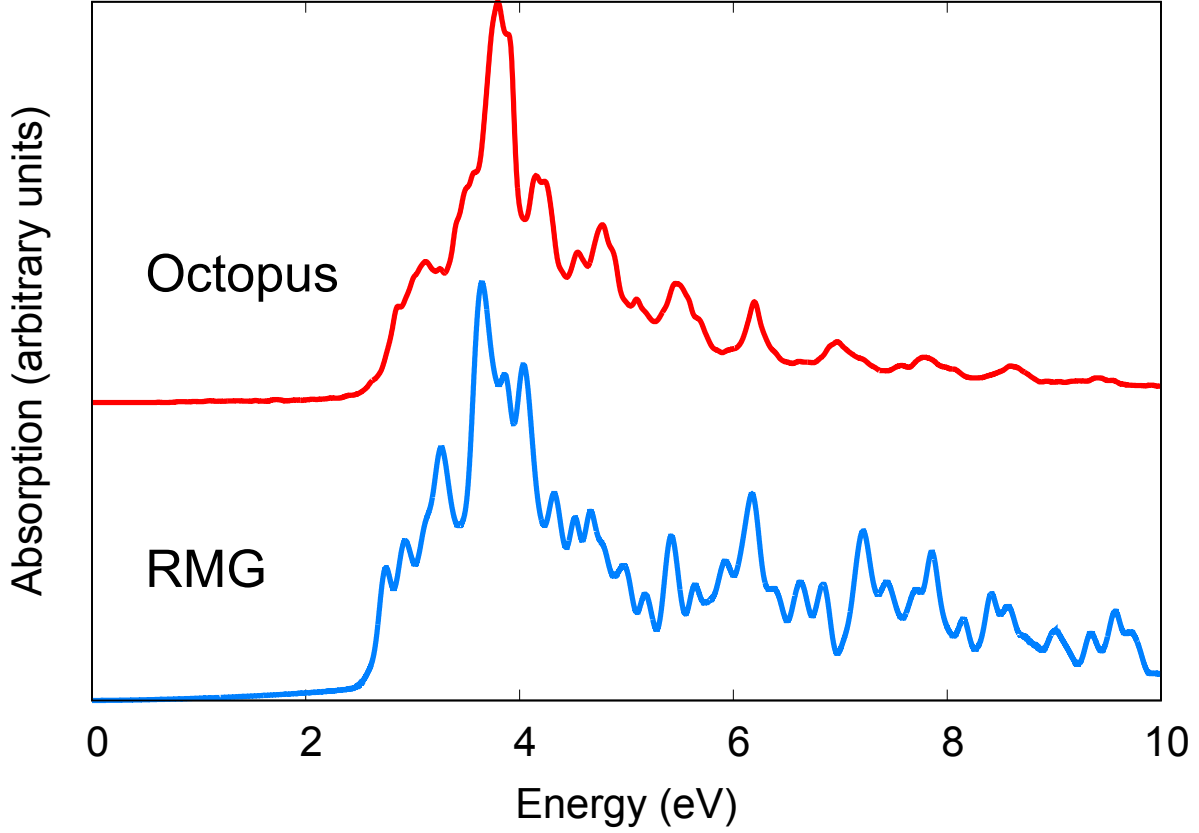


Figure 9: [Caption for Si optical conductivity figure. Should describe: $\sigma_{xx}(\omega)$ ($= \sigma_{yy} = \sigma_{zz}$ by cubic symmetry) from RMG compared with Octopus; indicate band gap region; optionally compare with experiment. If energy conservation panel included: describe it.]

with a common Hartree potential, yielding spin-resolved optical conductivities that can be compared with Octopus and with available experimental data.

[Computational details: - Unit cell: [describe geometry and magnetic ordering] - K-point sampling: $[N_k \times N_k \times N_k]$ - XC functional: [PBE or PBE+U; specify U if used] - Pseudopotential: [type, NCPP] - Initial spin configuration: ferromagnetic, moments on Cr sites - Time step and total simulation time: [values] - Broadening: [value] eV]

[Discussion points: - Agreement between RMG and Octopus for both spin channels - Physical interpretation: which spectral features are spin-up vs spin-down? Charge-transfer excitations, Cr d-d transitions - Spin current $\mathbf{J}_{\text{spin}} = \mathbf{J}^{\uparrow} - \mathbf{J}^{\downarrow}$ if computed (Fruit 4 from CONTEXT.md) - Comparison with available experimental optical data if relevant - Note: ECD/MCD spectra not computed in this work (magnetic perturbation formalism reserved for future publication)]

5.5 Large-Scale Demonstration: NV Center in Diamond

As a final demonstration, we apply the periodic spin-polarized RT-TDDFT implementation to the negatively charged nitrogen-vacancy (NV^-) center in diamond, one of the most studied

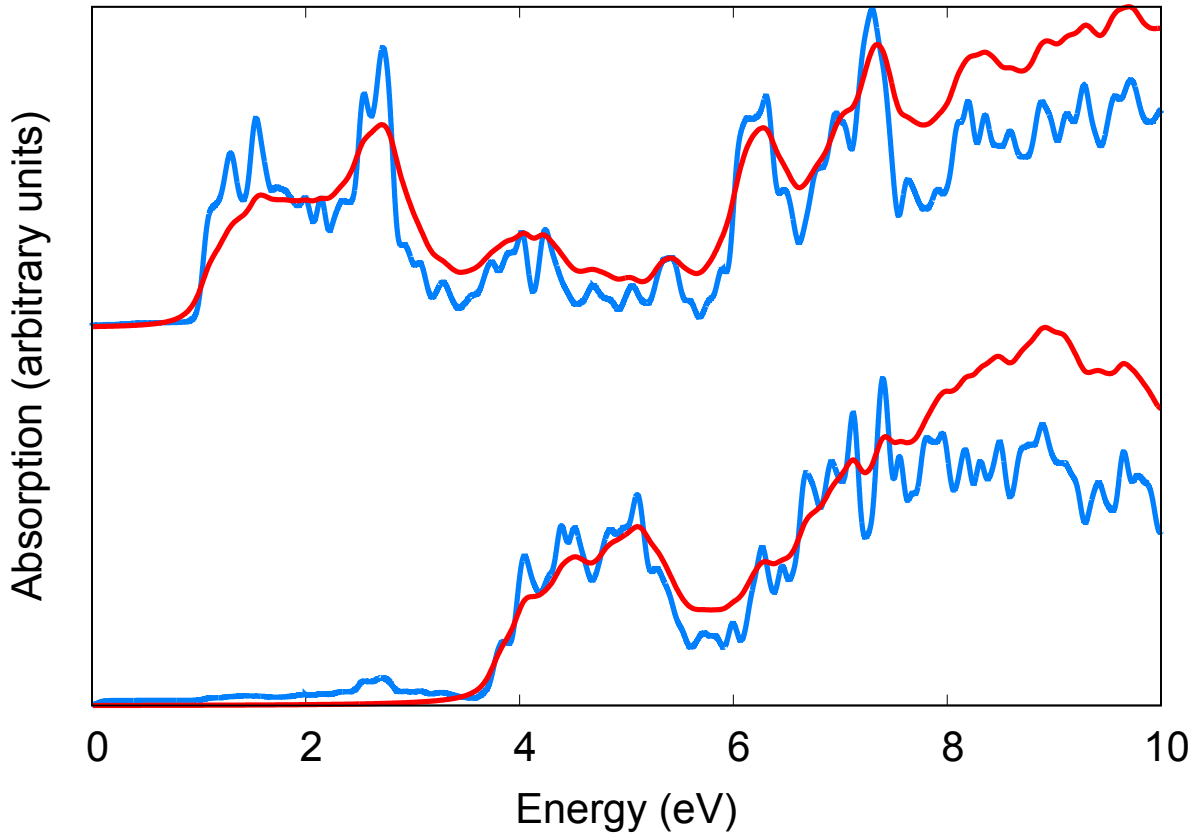


Figure 10: [Caption for CrI₃ optical conductivity figure. Should describe: spin-up and spin-down optical conductivities from RMG compared with Octopus; indicate the relevant spectral features (charge-transfer peaks, d-d transitions); note that spin channels are labeled consistently.]

point defects in solid-state physics.^{???} This system tests several capabilities simultaneously: spin-polarized real-time propagation, the optical response of a localized defect in a three-dimensional periodic host, convergence toward the dilute-defect limit, and computational scaling to thousands of atoms.

The NV⁻ center consists of a substitutional nitrogen atom adjacent to a carbon vacancy, oriented along a $\langle 111 \rangle$ crystallographic direction, with one additional trapped electron. Its ground state is an optically addressable spin triplet, with defect molecular orbitals located within the diamond band gap.^{???} Its long spin coherence time at room temperature makes it central to nanoscale magnetometry, quantum sensing, single-photon emission, and quantum information processing.^{?????}

In a standard periodic DFT or RT-TDDFT calculation the NV⁻ center is represented by a single defect in a periodically repeated supercell. The physical properties of the *isolated* defect are recovered only once the supercell is large enough that the interaction between the defect and its periodic images is negligible; identifying this convergence threshold is both a practical necessity and a physically meaningful result, since it establishes the minimum defect concentration at which the isolated-defect limit is reached.

The purpose of our demonstration is not to provide a complete many-body description of

the NV^- photophysics, which involves vibronic effects, singlet-triplet manifolds, and charge-state physics, but to demonstrate large-scale spin-polarized real-time simulations of optical response for increasing supercell size and defect concentration.

5.5.1 Electronic Structure and Optical Selection Rule

The NV^- triplet ground state (3A_2) has the single-particle configuration $a_1^2 e^1 e^1$: a doubly-occupied, non-degenerate a_1 orbital located primarily on the nitrogen atom with contributions from the three neighboring carbon atoms, and two singly-occupied, degenerate e -symmetry orbitals formed from those same three carbons. Both a_1 and e lie within the diamond band gap. Under the C_{3v} point group of the defect, the two-electron product decomposes as $E \times E = A_1 + A_2 + E$, so the lowest excited configuration obtained by promoting one electron from a_1 into e is itself doubly degenerate, matching the well-known 3E excited state. Concretely, the two microstates spanning the 3E manifold correspond to promoting the minority-spin ($\downarrow$) electron out of a_1 into either of the two degenerate e orbitals; the resulting $^3A_2 \rightarrow ^3E$ transition is the origin of the well-known zero-phonon line (ZPL) at 1.94 eV.[?]

Because E transforms as the in-plane representation (x, y) under C_{3v} (with the C_3 axis along the defect's $\langle 111 \rangle$ axis), the $^3A_2 \rightarrow ^3E$ transition dipole is confined to the plane perpendicular to $\langle 111 \rangle$: it is symmetry-forbidden for light polarized along $\langle 111 \rangle$ and allowed for any polarization within the perpendicular plane. We tested this directly by applying the RT-TDDFT delta-kick perturbation along three directions: $[111]$ (parallel to C_3), and $[1\bar{1}0]$ and $[11\bar{2}]$ (both perpendicular to C_3). The ZPL feature is absent for the $[111]$ kick and present, at essentially the same strength, for both perpendicular directions, with the residual signal along $[111]$ smaller by roughly a factor of 50—consistent with, and substantially exceeding, the group-theory prediction. The ZPL feature further appears predominantly in the spin-down channel, exactly as expected from the $a_1(\downarrow) \rightarrow e(\downarrow)$ assignment above: the spin-up manifold has no corresponding low-lying transition available at this energy, since both spin-up e orbitals are already singly occupied in the ground state. [\[Figure placeholder: three-panel polarization comparison, \$\[111\]\$ vs. \$\[1\bar{1}0\]\$ vs. \$\[11\bar{2}\]\$, spin-up/spin-down resolved. Correct the axis-label parallel/perpendicular assignment before finalizing the source figure—the underlying data are correct, only the descriptive labels need fixing.\]](#)

This result is independent of the supercell-size and active-space convergence questions addressed below: it follows from the defect's point-group symmetry alone, and therefore validates the implementation's velocity-gauge, polarization-resolved response independently of any of the numerical convergence caveats that follow. [\[Optional: brief note/plots of the \$a_1\$ and \$e\$ defect-orbital isosurfaces and/or spin density, if included.\]](#)

5.5.2 Linear-Response Optics (LOPTICS): An Independent Ground-State Cross-Check

Throughout the remainder of this demonstration we cross-validate our RT-TDDFT spectra against RMG's ground-state linear-response optics module (LOPTICS), which evaluates the imaginary part of the dielectric tensor directly from a sum over Kohn-Sham states, with no

time propagation involved. In atomic units,

$$\epsilon_2^{\alpha\beta}(\omega) = \frac{4\pi^2}{\Omega\omega^2} \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{v,c} (f_{v\mathbf{k}} - f_{c\mathbf{k}}) \mathcal{J}_{vc}^{\alpha}(\mathbf{k}) [\mathcal{J}_{vc}^{\beta}(\mathbf{k})]^* \delta(E_{c\mathbf{k}} - E_{v\mathbf{k}} - \hbar\omega), \quad (19)$$

where Ω is the unit-cell volume, $w_{\mathbf{k}}$ the $\mathbf{k}$ -point weight, $f_{n\mathbf{k}}$ the occupation of state n at $\mathbf{k}$, and the sum runs over occupied (v) and unoccupied (c) Kohn-Sham states. The real part $\epsilon_1(\omega)$ is reconstructed afterward via a Kramers-Kronig transform, and the energy-conserving δ -function is represented numerically by a Gaussian, re-broadened with a Lorentzian whose width sets the effective spectral resolution.

Critically, the matrix element $\mathcal{J}_{vc}^{\alpha}(\mathbf{k})$ entering Eq. (19) is *identical* to the current-operator matrix element defined in Eq. (??) (Sec. ??) and used throughout the real-time propagation: it includes the same nonlocal-pseudopotential correction, computed by the same underlying routine, not a bare momentum matrix element. LOPTICS and RT-TDDFT therefore start from exactly the same physical ingredients—ground-state Kohn-Sham orbitals and the same gauge-consistent current operator—and differ only in how the optical response is extracted from them: LOPTICS by direct sum-over-states evaluation in the frequency domain, RT-TDDFT by explicit real-time propagation followed by a Fourier transform. Because LOPTICS’s frequency resolution is set only by the chosen broadening, rather than by a finite propagation time, it provides an essentially exact reference against which the resolution-limited real-time spectra can be validated, as used throughout the rest of this section. No local-field correction is included at this level, consistent with an independent-particle treatment of the response.

5.5.3 Convergence with Supercell Size and Brillouin-Zone Sampling

We investigate the approach to the isolated-defect limit using a matrix of diamond supercells spanning three sizes and, for the two smaller sizes, two Brillouin-zone samplings each: a 64-atom supercell at both Γ and a $4 \times 4 \times 4$ Monkhorst-Pack mesh; a 512-atom supercell at both Γ and a $2 \times 2 \times 2$ mesh; and a 4096-atom supercell at Γ only, each containing a single NV^- center.

Table 3: Summary of the supercell-size/ k -sampling test matrix. RT-TDDFT results are cross-validated by LOPTICS (Sec. 5.5.2) at every entry; agreement is close except where noted.

System	k -mesh	Result
64 atoms	Γ	single dominant peak, ~ 1.9 eV
64 atoms	$4 \times 4 \times 4$	multiplet, ~ 1.4 – 2.4 eV
512 atoms	Γ	single peak, ~ 1.93 eV
512 atoms	$2 \times 2 \times 2$	[RT-TDDFT currently shows a single peak at short propagation time; re-running]
4096 atoms	Γ	single dominant peak; position depends on active-space size (Sec. 5.5.4)

At fixed cell size, it is moving from Γ -only to a dense k -mesh that reveals the multiplet structure at 64 atoms, not the reverse. This is an important point for interpreting the table: a clean, single-peak spectrum obtained at Γ in a small cell is *not*, by itself, evidence that the defect is behaving as an isolated emitter—it only means that a single point of what may still

be a dispersive, defect-coupled band has been sampled. Only when the spectrum becomes insensitive to k -sampling, as it does between 64 and 512 atoms, can convergence toward the isolated-defect limit be claimed with confidence. By this criterion, LOPTICS indicates that even the 512-atom cell has not fully reached that limit: at $2 \times 2 \times 2$ sampling it resolves a split peak that is not visible in the Γ -only result at the same size, implying non-negligible residual defect-image coupling even at 512 atoms. [State the resolution once the longer RT-TDDFT run at 512 atoms/ $2 \times 2 \times 2$ completes: either (a) RT-TDDFT recovers the same splitting once propagated long enough, confirming the coupling is real and that the earlier single-peak appearance was a finite-propagation-time resolution artifact, or (b) the methods genuinely disagree, which would need further investigation.]

An equivalent, and computationally more natural, way to probe the 512-atom/ $2 \times 2 \times 2$ result is via the real-space supercell-folding equivalence: a $k \times k \times k$ Monkhorst-Pack sampling of a cell is formally equivalent to a Γ -only calculation on the k^3 -fold real-space replica of that cell, provided the replica’s self-consistent solution respects the same periodicity as the original cell (i.e., no explicit symmetry-breaking to a larger magnetic unit cell). The $2 \times 2 \times 2$ real-space replica of the 512-atom cell is precisely the 8-defect, 4096-atom system used in Sec. 5.5.5 below; propagating that system at Γ only, in its translationally-simple (ferromagnetic) configuration, provides a second, independent route to the same 512-atom/ $2 \times 2 \times 2$ answer, without the added cost and complexity of explicit non- Γ real-time propagation. [Confirm numerically that the Γ -only, FM-configuration result for the 8-defect/4096-atom system matches the direct 512-atom/ $2 \times 2 \times 2$ result, as the folding argument predicts.]

[Computational details: - Host: diamond cubic, lattice constant [a] Å - Defect: single NV⁻ center (negatively charged) - Spin state: triplet ground state ($S = 1$) - XC functional: [value]; Pseudopotential: [value] - Active space (production, total states): 64 atoms: $N_{\text{basis}} = 13,640$ ($\approx 213/\text{atom}$); 512 atoms: $N_{\text{basis}} = 13,696$ ($\approx 27/\text{atom}$); 4096 atoms: $N_{\text{basis}} = 13,696$ ($\approx 3.3/\text{atom}$)—see Sec. 5.5.4 for the convergence implications of this shrinking per-atom ratio - Time step and total propagation time: [values] - Computational platform: Frontier at ORNL; nodes used: [values per system size]]

[Figure: RT-TDDFT and LOPTICS spectra overlaid for each row of Table 3, on a common energy axis. Label the ZPL region and, for the 64-atom/ $4 \times 4 \times 4$ case, the multiplet envelope.]

Figure 11: [Caption describing the convergence behavior and identifying the minimum supercell size at which the isolated-defect spectrum is recovered, per the k -sensitivity criterion above.]

5.5.4 Active-Space Convergence: Virtual Orbitals and the Occupied-State Window

Reaching the 4096-atom system size requires reducing the active space—the set of Kohn-Sham states actually propagated in real time—well below what is used for the smaller systems, purely for computational cost. Two independent truncations are available: the number of virtual (initially unoccupied) states added above the physically occupied manifold (the RMG input key `unoccupied_states_per_kpoint`, denoted N_{virt} below—not N_{occ} as in our working notes, since it is the virtual-state count that is varied), and the index of the lowest

occupied state included in the propagation (`tddft_start_state`), below which states are frozen at fixed occupation and do not contribute to the time-dependent density. The total active-space size achievable within a fixed compute budget is comparable across system sizes ($N_{\text{basis}} \approx 13,600\text{--}13,700$ states, Sec. 5.5.3), so the resulting per-atom active space falls sharply with system size: roughly 213 states/atom at 64 atoms, 27 states/atom at 512 atoms, and only 3.3 states/atom at 4096 atoms.

Independently varying `tddft_start_state` at fixed N_{virt} , and N_{virt} at fixed occupied window, at 512 atoms shows that both truncations shift the computed ZPL, with the virtual-side truncation producing the larger effect. Truncating the occupied window at fixed $N_{\text{virt}} = 3200$ moves the peak from ≈ 1.93 eV at `tddft_start_state` = 0—in close agreement with the experimental value of 1.94 eV—to ≈ 2.05 eV at `tddft_start_state` = 800 (of 1024 valence states); the corresponding N_{virt} sweep at fixed occupied window produces a substantially larger shift over a comparable range of active-space reduction. Both trends are the expected signature of a variationally under-converged excited state, not a numerical artifact: in the linear-response regime, RT-TDDFT is equivalent to the poles of the same response function obtained from a Casida-type eigenvalue problem in the same active space, and enlarging a *nested* particle-hole excitation space—exactly what increasing N_{virt} or decreasing `tddft_start_state` does—can only improve, never worsen, each ordered approximate root (a Hylleraas–Undheim/MacDonald-type interlacing result, the same principle behind the monotonic improvement of a truncated-CI excited-state energy with basis size). The same direction of shift is seen at 4096 atoms across active-space settings spanning $N_{\text{virt}} = 4000\text{--}9000$, confirming the trend persists at the flagship system size.

The production active-space sizes used for the size-convergence results in Sec. 5.5.3 are $N_{\text{basis}} = 13,640$, 13,696, and 13,696 for the 64-, 512-, and 4096-atom systems respectively—a comparable total active-space size across all three, rather than a fixed per-atom ratio, reflecting the largest active space practically feasible within a common compute budget at each size. [\[Report the resulting ZPL peak position at these final production settings for each system size, superseding the exploratory values quoted above for illustrating the convergence trend.\]](#)

Because the resulting per-atom active space is far smaller at 4096 atoms (3.3 states/atom) than at 512 atoms (27 states/atom), the larger deviation of the flagship system’s peak position from the experimental ZPL is most plausibly dominated by active-space truncation rather than by residual defect-image coupling—the latter having already been shown, in Sec. 5.5.3, to become small well before 4096 atoms. This mirrors the active-space convergence study already performed for finite systems in paper 1 (benzene, C_{60} , and Ag nanorods), extended here to a periodic defect with an additional, occupied-side truncation knob that the finite-system case did not require. Sec. 5.5.5 below provides a direct, controlled test of this attribution: by comparing the 1-, 8-, and 64-defect systems at the *same* total size, and therefore the *same* N_{virt} , any peak-position difference between them isolates the concentration/defect-image-coupling effect specifically, with active-space truncation held fixed as a common systematic across all three. [\[Figure placeholder: peak position vs. \$N_{\text{virt}}\$ \(at 512 and 4096 atoms\) and vs. `tddft\_start\_state` \(at 512 atoms\), showing the monotonic approach toward the experimental ZPL.\]](#) [\[If feasible, report an extrapolated or higher- \$N_{\text{virt}}\$ result for the 4096-atom system, using the 512-atom active-space sweep as a calibration for the expected remaining shift.\]](#)

5.5.5 Concentration-Matched Comparison: Ferromagnetic versus Antiferromagnetic Defect Ordering

The supercell-size convergence study above varies defect concentration and total simulation-cell size together: each cell contains exactly one NV center, so lowering the defect density is only possible by enlarging the cell. To isolate the effect of defect concentration from that of absolute cell size, we make further use of the 8-defect, 4096-atom replica introduced in Sec. 5.5.3, and construct one additional, more concentrated companion system: a $4 \times 4 \times 4$ real-space replica of the 64-atom (single-defect) cell, yielding 64 NV centers in a 4096-atom host. Together with the original single-defect 4096-atom calculation, these three systems—64, 8, and 1 NV centers—share an identical simulation-cell size and real-space grid, differing only in defect concentration, by two orders of magnitude. Crucially, because all three share the same total atom count, they can also be propagated with the *same* active-space size N_{virt} —unlike the direct single-defect size series in Sec. 5.5.3, where N_{virt} per atom necessarily shrinks as the cell grows. This is a deliberate design choice rather than an incidental convenience: it allows any peak-position shift among the 1-, 8-, and 64-defect results to be attributed to defect concentration specifically, rather than to the confound between active-space truncation and system size identified in Sec. 5.5.4. All three systems use $N_{\text{virt}} = 8200$ (≈ 2 virtual orbitals per carbon atom) and `tddft_start_state= 2700`, a deliberate compromise between the two 4096-atom active-space choices discussed in Sec. 5.5.4: closer to full occupancy than the $N_{\text{virt}} = 9000/\text{tddft_start_state}= 4100$ setting, but with a substantially larger virtual space than the initial $N_{\text{virt}} \approx 4000/\text{tddft_start_state}= 0$ setting. [Confirm: is the 1-defect entry in this comparison a fresh calculation at these compromise settings, or is it identical to one of the two 4096-atom results already reported in Sec. 5.5.4? If the former, the two should be distinguished explicitly rather than both called “the 4096-atom single-defect result.”]

At these densities the individual $S = 1$ spin-triplet defect moments are no longer independent, and the relevant physical question becomes how they order relative to one another. We consider two limiting configurations: a ferromagnetic (FM) arrangement, in which all defect spins are aligned parallel, and an antiferromagnetic (AFM) arrangement, in which neighboring defect moments alternate in sign across the defect lattice. The total-energy difference between these two configurations,

$$\Delta E_{\text{FM-AFM}} = E_{\text{FM}} - E_{\text{AFM}}, \quad (20)$$

provides an estimate of the effective magnetic exchange coupling between NV centers mediated through the diamond host, analogous to exchange constants extracted from broken-symmetry DFT calculations of multi-radical or polynuclear transition-metal systems.[cite: broken-symmetry / Heisenberg exchange reference]

Preliminarily, we find the ferromagnetic configuration to be energetically favored over the antiferromagnetic configuration *at both defect concentrations* (8- and 64-defect systems), with $\Delta E_{\text{FM-AFM}} = [\text{value, meV or meV/defect}]$ for the 64-defect system and $[\text{value}]$ for the 8-defect system. [Comment on whether the coupling strengthens or weakens with increasing defect separation (i.e. between the 64- and 8-defect cases), consistent with a superexchange- or wavefunction-overlap-mediated mechanism rather than long-range dipolar coupling.]

We have additionally computed the linear optical response (LOPTICS, see Sec. 5.5.2) for both the FM and AFM configurations at both defect concentrations, providing a converged baseline against which the corresponding real-time spectra can be validated once available. [Figure placeholder: LOPTICS spectra for FM vs. AFM configurations, 64- and 8-defect systems, compared against the single-defect (1-defect) 4096-atom reference spectrum from Fig. 11. Discuss whether magnetic ordering produces a resolvable spectral signature, or whether the FM and AFM optical responses are nearly indistinguishable at these concentrations.] [RT-TDDFT real-time spectra for the FM and AFM systems: still running at time of writing—insert once complete, using the LOPTICS results above as the reference for judging active-space and propagation-length convergence.]

We note that we have not performed an analogous Γ -versus- $2 \times 2 \times 2$ LOPTICS check on the single-defect 4096-atom reference system used as the 1-defect baseline above. [State explicitly: based on the trend established in the main convergence study (defect-image coupling decreasing monotonically with cell size/increasing real-space defect separation), we do not expect qualitative changes at this size; a dense-k check is left for future work, e.g. in response to reviewer request.]

This comparison illustrates a further capability unique to a massively-scalable real-time implementation: beyond recovering the isolated-defect limit for a single center, RMG can directly address ensembles of interacting quantum defects at densities relevant to real NV-diamond samples, opening a path toward studying emergent collective phenomena—magnetic ordering and defect-defect optical coupling—that are inaccessible to small-cell approaches.

6 OpenAI

6.1 Large-Scale Demonstration: NV^- Center in Diamond

As a final demonstration, we apply the periodic spin-polarized RT-TDDFT implementation to the negatively charged nitrogen-vacancy center (NV^-) in diamond.^{???} This system combines several aspects of the present implementation: a localized defect embedded in a three-dimensional periodic host, spin-polarized electronic structure, polarization-resolved optical response, k -point sampling, and large active density-matrix propagation. The NV^- center also provides a stringent practical test because an isolated point defect must be represented by a sufficiently large supercell to reduce interactions with its periodic images.

The purpose of this section is not to provide a complete quantitative description of NV^- photophysics. In particular, the experimental zero-phonon line (ZPL) involves vibronic structure, excited-state relaxation, and electron–phonon coupling, none of which are included in the fixed-ion calculations considered here. Instead, the present calculations examine the electronic optical response in the energy region of the experimental ZPL and use the NV^- center to demonstrate how polarization selection rules, periodic defect-image interactions, and active-space truncation affect large-scale periodic RT-TDDFT calculations.

6.1.1 Defect electronic structure and polarization selection rules

The NV^- center consists of a substitutional nitrogen atom adjacent to a carbon vacancy, with one additional electron trapped at the defect.^{???} The defect axis is oriented along the crystallographic $[111]$ direction, which defines the C_3 symmetry axis of the approximately C_{3v} local defect environment. The defect states are located within the diamond band gap. The lower a_1 -like defect orbital has substantial nitrogen character together with contributions from the three carbon atoms adjacent to the vacancy, whereas the nearly degenerate e manifold is primarily localized on these three carbon atoms.

Within the collinear spin-polarized Kohn–Sham description, the triplet ground state associated with the 3A_2 manifold has the defect-state occupation

$$a_1^2 e_x^1 e_y^1. \quad (21)$$

In the spin representation used here, the lowest defect-associated optical excitation is dominated by promotion of a minority-spin electron from the occupied a_1 -like defect orbital into the unoccupied e manifold. This excitation is associated with the electronic transition underlying the experimentally observed $^3A_2 \rightarrow ^3E$ optical transition, whose ZPL occurs at approximately 1.945 eV.[?] The experimental ZPL energy is used below only as a reference for the spectral region of interest.

The polarization dependence of this transition provides a direct symmetry test for the periodic RT-TDDFT implementation. We considered a kick parallel to the NV symmetry axis,

$$\mathbf{e}_{\parallel} \parallel [111], \quad (22)$$

and two independent transverse directions,

$$\mathbf{e}_{\perp,1} \parallel [1\bar{1}0], \quad \mathbf{e}_{\perp,2} \parallel [11\bar{2}]. \quad (23)$$

[Confirm that the three applied kick directions were unit-normalized before comparing relative intensities.] For an approximately C_{3v} -symmetric NV^- center, the dipole operator parallel to the defect axis transforms as A_1 , whereas its two transverse components transform as E . Consequently, the $^3A_2 \rightarrow ^3E$ transition is electric-dipole allowed for polarization perpendicular to the $[111]$ axis and forbidden for polarization along that axis.

The calculated response follows this selection rule. The defect-associated feature in the experimental ZPL energy region is absent, within the numerical resolution of the calculation, for a kick applied parallel to $[111]$, whereas it is present for both transverse polarizations. The close correspondence between the two transverse responses provides an additional check that the local defect symmetry is retained in the simulation. This result validates the ability of the periodic velocity-gauge formulation to recover polarization-resolved optical selection rules for a localized spin-polarized defect.

[Insert Figure ?? here. Suggested content: (a) diamond supercell with the N atom, vacancy, and $[111]$ defect axis indicated; (b) isosurfaces of the occupied a_1 -like defect state and one representative member of the e doublet; (c) optional spin-density isosurface; and (d) polarization-resolved spectra for $\mathbf{e}_{\parallel}$, $\mathbf{e}_{\perp,1}$, and $\mathbf{e}_{\perp,2}$. Mark the experimental 1.945 eV ZPL only as a vertical reference line.]

6.1.2 Comparison between RT-TDDFT and independent-particle optical spectra

We next compare the real-time spectra with independent-particle optical spectra obtained using the LOPTICS implementation. The latter is based on a sum over Kohn–Sham transitions between occupied and unoccupied states of the ground-state electronic structure. Thus, LOPTICS provides a computationally less expensive linear-optical reference that resolves the transition structure of the same periodic defect array without real-time propagation of the time-dependent density matrix.

The comparison is particularly informative for the smallest NV^- -containing diamond cell, denoted below as $\text{NV-}x2$, which contains 64 atoms and is sampled using a $4 \times 4 \times 4$ k -point mesh. In this case, both RT-TDDFT and LOPTICS resolve several features in the sub-gap defect-transition region. The close agreement between the two approaches in the positions and overall structure of these features demonstrates that the periodic RT-TDDFT implementation reproduces the optical response expected from the corresponding ground-state Kohn–Sham transition manifold.

This agreement should not be interpreted as a comparison to the isolated-defect experimental spectrum. Both calculations describe the same finite-concentration periodic array of NV^- centers and inherit the same underlying approximate exchange-correlation functional, pseudopotentials, supercell geometry, and defect-image interactions. Instead, the agreement establishes that the real-time current response correctly reproduces the linear optical structure of the periodic system at the same electronic-structure level.

[Insert Figure ?? here. Suggested content: transverse-polarization RT-TDDFT and LOPTICS spectra for $\text{NV-}x2$ with $4 \times 4 \times 4$ sampling. A second panel or inset may show the corresponding $\text{NV-}x4$ result with $2 \times 2 \times 2$ sampling. Use the same broadening, energy range, and physically meaningful normalization for each comparison.]

6.1.3 Periodic defect arrays and approach toward the dilute-defect regime

A periodic calculation of a point defect describes an infinite array of periodically repeated defects rather than a strictly isolated defect. The resulting defect-derived states can retain non-negligible dispersion when the defect concentration is high enough for the defect orbitals, electrostatic fields, or host-mediated interactions to couple across the periodic cell. Sampling multiple k points does not create this coupling; rather, it resolves the k -dependent transition energies and oscillator strengths of the finite-concentration periodic defect array.

This effect is apparent from the contrast between Γ -point and finite- k calculations. For the 64-atom $\text{NV-}x2$ cell, the $4 \times 4 \times 4$ spectrum exhibits a pronounced multipeak structure in the defect-transition region, whereas the corresponding Γ -point calculation samples only one component of this periodic-defect band structure. The intermediate $\text{NV-}x4$ cell, containing 512 atoms, shows reduced but still detectable structure when sampled with a $2 \times 2 \times 2$ mesh. These results indicate that the defect-related transition remains sensitive to Brillouin-zone sampling even at the 512-atom scale and that periodic-image coupling is not yet negligible at this defect concentration.

To examine this behavior from a real-space perspective, we constructed explicitly replicated supercells having approximately the same total diamond volume as the single-defect

4096-atom NV- x 8 system. The NV- x 2 cell was replicated $4 \times 4 \times 4$, producing a 4096-atom cell containing 64 NV $^-$ centers, while the NV- x 4 cell was replicated $2 \times 2 \times 2$, producing a 4096-atom cell containing 8 NV $^-$ centers. These replicated systems preserve the geometry, grid spacing, lattice vectors, and local defect structure of the respective parent cells, while changing the real-space representation of the periodic defect array. Together with the 4096-atom NV- x 8 cell containing one defect, they provide a controlled sequence with 64, 8, and 1 NV $^-$ centers in comparable overall volumes.

Table 4: Real-space representations used to examine defect concentration and periodic-image coupling. NV- x 2 and NV- x 4 denote the 64- and 512-atom parent cells, respectively. The replicated calculations retain parallel collinear alignment of corresponding defect moments, thereby preserving the translational symmetry of the parent periodic-defect array.

System	Parent-cell replication	Total atoms	NV $^-$ centers
NV- x 2 k -sampled array	$4 \times 4 \times 4$ k mesh	64	1
NV- x 2 replicated array	$4 \times 4 \times 4$ real-space cell	4096	64
NV- x 4 k -sampled array	$2 \times 2 \times 2$ k mesh	512	1
NV- x 4 replicated array	$2 \times 2 \times 2$ real-space cell	4096	8
NV- x 8 dilute-defect cell	Γ point	4096	1

For a complete one-to-one correspondence between a k -sampled parent cell and its explicitly replicated Γ -point supercell, the large supercell would need to retain the full folded manifold of occupied and unoccupied states represented in the parent-cell calculation. For the present systems, this requirement would involve tens of thousands of unoccupied states and is computationally impractical. We therefore do not claim exact equivalence of the full optical spectra between the k -sampled and replicated descriptions. Instead, the replicated calculations are used to analyze the low-energy defect-transition region and to establish how the optical response evolves as the density of periodically repeated NV $^-$ centers is reduced.

[Insert Figure ?? here. Suggested content: spectra in the 1.2–3.0 eV defect-transition region for the 64-defect, 8-defect, and one-defect 4096-atom systems. Show one panel with the response per simulation volume and a second panel normalized per NV $^-$ center, if both normalizations are useful. The k -sampled parent-cell spectra may be included as dashed reference curves, but the caption should state explicitly that identical virtual manifolds were not retained in the large replicated cells.]

6.1.4 Active-space convergence and large-scale propagation

The computational cost of density-matrix RT-TDDFT grows rapidly with the number of propagated Kohn–Sham states. For each spin channel, the dimension of the propagated Hamiltonian and density matrices is approximately

$$N_{\text{basis}}^{\sigma} = N_{\text{occ}}^{\text{act},\sigma} + N_{\text{virt}}^{\sigma}, \quad (24)$$

where $N_{\text{occ}}^{\text{act},\sigma}$ is the number of occupied states retained in the active propagation manifold and N_{virt}^{σ} is the number of included unoccupied states. In the present discussion, the code

input keyword `Nocc` is denoted by N_{virt} , because it controls the number of unoccupied states retained above the occupied manifold.

To establish the numerical consequences of this truncation, we performed active-space tests for the 512-atom NV- $x4$ system. The option `tddft_start_state` was varied to reduce the number of occupied states participating in the real-time propagation, and N_{virt} was varied independently. In both cases, restricting the active manifold produces a blue shift of the defect-associated optical feature. Increasing either the active occupied manifold or the number of included virtual states lowers the peak toward a stable value over the range examined. This trend is reported here as an empirical active-space convergence result for the present calculations; it is not interpreted as a general variational principle for real-time TDDFT excitation energies.

The active-space tests provide essential context for the largest calculations. The 4096-atom NV- $x8$ calculation requires a restricted but computationally feasible active manifold in order to make spin-polarized real-time propagation practical. Consequently, the large-cell spectrum should be interpreted as evidence of the large-system periodic response and reduced defect concentration, while its absolute transition energy retains an active-space convergence uncertainty. In particular, a difference between the 4096-atom spectrum and the smaller-cell spectra cannot be assigned exclusively to suppression of defect-image coupling unless the active-space dependence is considered simultaneously.

[Insert Figure 4 here. Suggested content: (a) defect-region peak position versus N_{virt} for the 512-atom NV- $x4$ system; (b) peak position versus `tddft_start_state`, or equivalently versus $N_{\text{occ}}^{\text{act}}$; and (c) wall time per RT-TDDFT step versus $N_{\text{basis}}^{\sigma}$. Add the active-space choice used for the 4096-atom calculation as a vertical reference line.]

The largest calculations demonstrate that the periodic spin-polarized RT-TDDFT implementation can propagate defect-containing diamond cells with more than four thousand atoms and active-state manifolds containing several thousand Kohn–Sham orbitals. This capability makes it possible to move beyond the highly concentrated defect arrays implicitly represented by conventional small-supercell calculations and to examine the crossover toward the dilute-defect regime using first-principles real-time electron dynamics.

[Optional Supporting Information analysis: compare parallel and antiparallel collinear arrangements of defect moments in the replicated NV- $x2$ and NV- $x4$ arrays. Report $\Delta E_{\text{AP-FM}} = E_{\text{AP}} - E_{\text{FM}}$ per defect, together with spin-density plots, only after confirming that the local moments remain localized at the intended NV $^-$ sites. This analysis should be described as a diagnostic of exchange coupling within concentrated periodic defect arrays, not as a prediction of magnetic ordering for dilute NV centers.]

7 Summary and Outlook

[To be written after all other sections are complete. Should summarize the key contributions, state the validated capabilities, and outline planned extensions: (1) continuous driving field / nonlinear response, (2) non-collinear spin RT-TDDFT, (3) Ehrenfest dynamics combined with spin, (4) ultrasoft pseudopotentials.]

7.1 REMAINING STUFF

To describe the interaction of electrons with an external electromagnetic field, we start from the time-dependent Kohn–Sham (TDKS) equation for the occupied Kohn–Sham orbitals,

$$i\frac{\partial}{\partial t}\psi_n(\mathbf{r}, t) = \hat{H}_{\text{KS}}(t)\psi_n(\mathbf{r}, t), \quad (25)$$

where, in atomic units, the TDKS Hamiltonian in the presence of external scalar and vector potentials is written as

$$\hat{H}_{\text{KS}}(t) = \frac{1}{2} \left[\hat{\mathbf{p}} + \frac{1}{c} \mathbf{A}_{\text{ext}}(\mathbf{r}, t) \right]^2 - \Phi_{\text{ext}}(\mathbf{r}, t) + \hat{V}_{\text{ion}} + \hat{V}_{\text{H}}[n](\mathbf{r}, t) + \hat{V}_{\text{xc}}[n](\mathbf{r}, t). \quad (26)$$

Here $\hat{\mathbf{p}} = -i\nabla$, $\Phi_{\text{ext}}(\mathbf{r}, t)$ and $\mathbf{A}_{\text{ext}}(\mathbf{r}, t)$ are the external scalar and vector potentials, and $\hat{V}_{\text{ion}}$, $\hat{V}_{\text{H}}[n]$, and $\hat{V}_{\text{xc}}[n]$ denote the ionic, Hartree, and exchange-correlation contributions, respectively. In practical implementations based on pseudopotentials, $\hat{V}_{\text{ion}}$ may contain both local and nonlocal parts. The corresponding electric and magnetic fields are obtained from

$$\mathbf{E}(\mathbf{r}, t) = -\nabla\Phi_{\text{ext}}(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathbf{A}_{\text{ext}}(\mathbf{r}, t)}{\partial t}, \quad \mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}_{\text{ext}}(\mathbf{r}, t). \quad (27)$$

For finite systems, such as atoms and molecules, the interaction with light is commonly treated within the long-wavelength, or electric-dipole, approximation. In this limit the spatial phase factor of the electromagnetic wave,

$$e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}, \quad (28)$$

varies negligibly over the extent of the system, so that $\mathbf{q}\cdot\mathbf{r} \ll 1$ and the field may be regarded as spatially uniform. The interaction is then conveniently written in the length gauge as

$$\hat{V}_{\text{ext}}^{(L)}(t) = -\mathbf{r} \cdot \mathbf{E}(t). \quad (29)$$

Equivalently, one may regard this perturbation as arising from the scalar potential

$$\Phi_{\text{ext}}^{(L)}(\mathbf{r}, t) = \mathbf{r} \cdot \mathbf{E}(t), \quad \mathbf{A}_{\text{ext}}^{(L)}(\mathbf{r}, t) = 0. \quad (30)$$

Within this approximation, the magnetic component of the radiation field is neglected, and the physically relevant quantity is the field polarization direction. Since the wavevector $\mathbf{q}$ is no longer retained explicitly, the terms transverse and longitudinal are not the most useful descriptors at this stage.

The length-gauge form of Eq. (29) is well suited for finite systems, but becomes problematic for periodic solids. In particular, the scalar potential $-\mathbf{E}(t) \cdot \mathbf{r}$ is not lattice periodic and is therefore incompatible with Bloch-periodic boundary conditions. In real-space representations this issue is often manifested as a discontinuous sawtooth-like potential across the cell boundary. More fundamentally, the response of a periodic solid to a homogeneous electric field is more naturally expressed in terms of macroscopic current or polarization rather than a direct scalar-potential coupling through the position operator.

For this reason, periodic RT-TDDFT is commonly formulated in the velocity gauge. For a spatially homogeneous electric field, a gauge transformation removes the scalar potential and introduces a purely time-dependent vector potential,

$$\Phi_{\text{ext}}^{(V)}(\mathbf{r}, t) = 0, \quad \mathbf{A}_{\text{ext}}^{(V)}(\mathbf{r}, t) = \mathbf{A}(t), \quad \mathbf{E}(t) = -\frac{1}{c} \frac{d\mathbf{A}(t)}{dt}. \quad (31)$$

The TDKS Hamiltonian then takes the form

$$\hat{H}_{\text{KS}}^{(V)}(t) = \frac{1}{2} \left[\hat{\mathbf{p}} + \frac{1}{c} \mathbf{A}(t) \right]^2 + \hat{V}_{\text{ion}} + \hat{V}_{\text{H}}[n](\mathbf{r}, t) + \hat{V}_{\text{xc}}[n](\mathbf{r}, t). \quad (32)$$

This formulation is gauge-equivalent to the dipole length gauge of Eq. (29), but is considerably better suited for periodic systems because the perturbation preserves translational symmetry. It is important to emphasize that the homogeneous-field velocity gauge of Eq. (31) is still a dipole-limit description. Since $\mathbf{A}(t)$ carries no spatial dependence, it does not represent a full propagating electromagnetic wave and does not generate a magnetic field, $\nabla \times \mathbf{A}(t) = 0$.

A more general description of radiative coupling is obtained by retaining the spatial dependence of the external field. In Coulomb gauge,

$$\nabla \cdot \mathbf{A}_{\text{ext}}(\mathbf{r}, t) = 0, \quad (33)$$

a transverse electromagnetic plane wave may be written as

$$\mathbf{A}_{\text{ext}}(\mathbf{r}, t) = \mathbf{A}_0 e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} + \text{c.c.}, \quad \mathbf{q} \cdot \mathbf{A}_0 = 0. \quad (34)$$

The condition $\mathbf{q} \cdot \mathbf{A}_0 = 0$ expresses the transverse character of the field, namely that the polarization direction is perpendicular to the propagation direction $\mathbf{q}$. In this case both electric and magnetic components are present, with

$$\mathbf{E}(\mathbf{r}, t) = -\frac{1}{c} \frac{\partial \mathbf{A}_{\text{ext}}(\mathbf{r}, t)}{\partial t}, \quad \mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}_{\text{ext}}(\mathbf{r}, t). \quad (35)$$

This finite- $\mathbf{q}$ formulation is more general than the dipole approximation because it explicitly retains momentum transfer and spatial phase information. In the long-wavelength limit, $e^{i\mathbf{q} \cdot \mathbf{r}} \approx 1$, it reduces to the homogeneous-field description discussed above.

It is useful to distinguish the transverse plane wave of Eq. (34) from a scalar-potential perturbation of the form

$$\Phi_{\text{ext}}(\mathbf{r}, t) = \Phi_0 e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}, \quad \mathbf{A}_{\text{ext}}(\mathbf{r}, t) = 0. \quad (36)$$

In this case the electric field is

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \Phi_{\text{ext}}(\mathbf{r}, t) = -i\mathbf{q} \Phi_0 e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}, \quad (37)$$

which is parallel to $\mathbf{q}$. Such a perturbation is therefore longitudinal and probes the density response of the system at finite momentum transfer. This type of field is natural for the

calculation of density-density response functions, loss spectra, and plasmonic excitations, whereas the transverse vector-potential form of Eq. (34) describes radiative coupling to an external electromagnetic wave.

The distinction between these external-field representations is also reflected in the analysis of RT-TDDFT simulations. For finite systems treated in the dipole approximation, the response is commonly characterized through the time-dependent dipole moment and its Fourier transform, which yield optical absorption spectra and frequency-dependent polarizabilities. In periodic systems formulated in the velocity gauge, the natural observable is instead the time-dependent current density, from which optical conductivity, dielectric response, and related quantities may be extracted. In the exact limit, length-gauge and velocity-gauge descriptions are equivalent, provided that observables and operators are transformed consistently. In practice, however, the current-based formulation is the natural choice for periodic systems under homogeneous fields, while the more general finite- $\mathbf{q}$ framework provides a clear route toward momentum-resolved response beyond the dipole limit.

In the present work we employ the homogeneous-field velocity-gauge formulation of Eqs. (31) and (32) as the working framework for periodic RT-TDDFT. This choice preserves periodic boundary conditions, provides direct access to current-based observables, and establishes a natural starting point for future extensions to finite- $\mathbf{q}$ perturbations and momentum-resolved spectroscopies.

8 Methodology

All equations in this section are given in atomic units ($\hbar = m_e = e = 4\pi\epsilon_0 = 1$) unless stated otherwise.

8.1 Overview of RMG

[Brief overview of RMG: real-space grid, multigrid acceleration, GPU scalability. Refer heavily to paper 1[?] and the RMG code paper[?] rather than repeating. Keep to 1–2 paragraphs. Focus on what is new or relevant for the periodic case: k-point sampling, Bloch states, periodic boundary conditions. Mention norm-conserving pseudopotentials (NCPP only).]

[Note from CONTEXT.md: the code uses norm-conserving PP only (asserted in RmgT-ddft.cpp line 287). Ultrasoft PP requires $S \neq 1$ in the propagation (eldyn_nonort.cpp path) and is not yet implemented. State this as an explicit scope limitation here.]

8.2 Interaction of Matter with Electromagnetic Radiation

[DERIVATION NEEDED from Jacek's LaTeX files. Content outline: - Start from the full minimal-coupling Hamiltonian with EM field treated as a plane wave: $\exp(i\omega t - i\mathbf{q} \cdot \mathbf{r})$ - Derive the long-wavelength (dipole) approximation: $\mathbf{q} \rightarrow 0$, i.e., $\lambda \gg$ system size, valid for optical/UV and unit-cell scales - Show that in this limit the vector potential $\mathbf{A}$ becomes spatially uniform: $\mathbf{A}(\mathbf{r}, t) \rightarrow \mathbf{A}(t)$ - Result: the two natural gauge choices in the long-wavelength limit are the length gauge (scalar potential $\Phi = -\mathbf{r} \cdot \mathbf{E}$, $\mathbf{A} = 0$) and the velocity gauge ($\Phi = 0$, $\mathbf{A}(t)$ uniform) - Physical interpretation: length gauge couples to position

(dipole), velocity gauge couples to momentum (current) - Reference: Mattiat & Luber (2022) Section 2.3; Yabana & Bertsch (1996)]

The time-dependent Kohn-Sham (TDKS) equations governing the electronic dynamics under an external electromagnetic perturbation take the form

$$i\frac{\partial}{\partial t}\psi_n(\mathbf{r}, t) = \hat{H}_{\text{KS}}(t) \psi_n(\mathbf{r}, t), \quad (38)$$

where $\hat{H}_{\text{KS}}(t)$ is the time-dependent Kohn-Sham Hamiltonian. [Expand: write out $\hat{H}_{\text{KS}}(t)$ with all terms including the EM coupling, then derive the long-wavelength limit.]

=====

8.3 Gauge Transformation: Length to Velocity Gauge

Content outline: - State clearly why length gauge fails for periodic systems: the position operator $\hat{\mathbf{r}}$ is not translationally invariant; dipole moment of a periodic system is multivalued (refer to modern theory of polarization: Resta 1994, King-Smith & Vanderbilt 1993) - Present the Göppert-Mayer gauge transformation explicitly: gauge function $\chi(\mathbf{r}, t) = \mathbf{r} \cdot \mathbf{A}(t)$, transformations of potentials and wavefunctions (Mattiat eqs. 15–19) - Show that the length gauge TDKS (eq. 39) transforms to the velocity gauge TDKS (eq. 40) - CRITICAL: the nonlocal pseudopotential $\hat{V}^{\text{nl}}$ does NOT commute with the gauge transformation. The gauge-transformed $\hat{V}^{\text{nl}}$ acquires an additional term (Mattiat eq. 23). Derive this explicitly here or refer to Appendix A. - Result: velocity gauge TDKS Hamiltonian with $\mathbf{A}(t)$ coupling - Reference: Mattiat & Luber (2022) Sections 2.3.1, Appendix A, B; Pickard & Mauri (2001)

In the length gauge, the TDKS equations read

$$i\frac{\partial}{\partial t}\psi_n(\mathbf{r}, t) = \left[\hat{H}_0 + \mathbf{r} \cdot \mathbf{E}(t) \right] \psi_n(\mathbf{r}, t), \quad (39)$$

where $\hat{H}_0$ is the unperturbed Kohn-Sham Hamiltonian and $\mathbf{E}(t)$ is the electric field. Applying the gauge transformation $\chi(\mathbf{r}, t) = \mathbf{r} \cdot \mathbf{A}(t)$, where the vector potential satisfies $\mathbf{E}(t) = -\partial\mathbf{A}/\partial t$, the velocity gauge TDKS equations take the form

$$i\frac{\partial}{\partial t}\psi_n(\mathbf{r}, t) = \left[\frac{1}{2} (\hat{\mathbf{p}} + \mathbf{A}(t))^2 + V(\mathbf{r}, t) + \tilde{V}^{\text{nl}}(\mathbf{r}, t) \right] \psi_n(\mathbf{r}, t), \quad (40)$$

where $\tilde{V}^{\text{nl}}$ denotes the gauge-transformed nonlocal pseudopotential.?? [Expand: write out all terms of $\tilde{V}^{\text{nl}}$ explicitly. Derive the linearization in $\mathbf{A}(t)$ that gives the perturbative coupling term. Connect to Appendix A.]

8.4 Vector Potential and Magnetic Perturbations

Content outline: - Show that the velocity gauge framework with $\mathbf{A}(t)$ naturally accommodates both electric and magnetic perturbations - Electric perturbation in the long-wavelength limit: $\mathbf{A}(t)$ spatially uniform, $\mathbf{B} = 0$ - Magnetic perturbation (symmetric gauge): $\mathbf{A}(\mathbf{r}) = \frac{1}{2}\mathbf{B} \times (\mathbf{r} - \mathbf{R}_G)$, spatially dependent, introduces gauge origin $\mathbf{R}_G$ - Note: magnetic per-

turbation sets up ECD/MCD as future extension (not implemented in present work but theoretically grounded here) - The gauge origin dependence with finite basis sets and the standard remedies (GIAO, IGLO) – brief mention, refer to Mattiat for details - Reference: Mattiat & Luber (2022) Section 2.3.2; Pickard & Mauri (2001)

[Note: This subsection establishes the theoretical foundation for magnetic perturbations. The current paper implements only electric perturbations (A(t) uniform). ECD/MCD calculations are future work. State this explicitly at the end of the subsection.]

8.5 Current Operator and Nonlocal Pseudopotential Contributions

This is the most important new subsection. Full derivation required. Content outline:

(a) Canonical vs mechanical momentum: - Canonical: $\hat{\mathbf{p}} = -i\nabla$ - Mechanical: $\hat{\boldsymbol{\pi}} = \hat{\mathbf{p}} + \mathbf{A}(t)$ - Velocity operator: $\hat{\mathbf{v}} = (1/i)[\hat{\mathbf{r}}, \hat{H}]$ - When $V^{\text{nl}} \neq 0$: $\hat{\mathbf{v}} \neq \hat{\mathbf{p}}/m$ – extra commutator term $[\hat{\mathbf{r}}, V^{\text{nl}}]$ - Ref: Starace (1971,1973); Mattiat & Luber eq.12

(b) Paramagnetic and diamagnetic current: - Full current: $\hat{\mathbf{J}} = \hat{\mathbf{J}}_{\text{para}} + \hat{\mathbf{J}}_{\text{dia}}$ - Paramagnetic: $\hat{\mathbf{J}}_{\text{para}} \propto \text{Re}[\psi^*(-i\nabla)\psi]$ - Diamagnetic: $\hat{\mathbf{J}}_{\text{dia}} \propto \mathbf{A}(t)|\psi|^2$ - $\mathbf{J}_{\text{dia}} = 0$ after $t=0$ in our perturbative scheme ($\mathbf{A}=0$ for $t>0$) - Justify this explicitly as valid in linear response regime

(c) Continuity equation: - $\partial\rho/\partial t + \nabla \cdot \mathbf{J} = 0$ - Use as consistency/sanity check for the current operator

(d) Current in density matrix formulation: - Macroscopic current: $J_\alpha(t) = \text{Re Tr}[\mathbf{P}(t) \cdot \mathbf{J}^\alpha]$ - Current operator matrix: $J_{ab}^\alpha = p_{ab}^\alpha + [\hat{r}_\alpha, V^{\text{nl}}]_{ab}$ - where $p_{ab}^\alpha = \langle \psi_a | (-i\nabla_\alpha + k_\alpha) | \psi_b \rangle$ (the k_α term comes from the Bloch theorem: see eq.XX) - This is what VecPHmatrix() + CurrentNlpp() compute in the code

(e) Nonlocal PP contribution $[r, V^{\text{nl}}]$: - For Kleinman-Bylander separable pseudopotentials: $V^{\text{nl}} = \sum_{I,lm} |\beta_I^{lm}\rangle h_{ij}^l \langle \beta_I^{lm}|$ - Compute $[\hat{r}_\alpha, V^{\text{nl}}]$ explicitly: $[\hat{r}_\alpha, V^{\text{nl}}]|\psi\rangle = \sum_{I,lm} |\beta_I^{lm}\rangle h_{ij}^l \langle [\hat{r}_\alpha, \beta_I^{lm}] | \psi \rangle$ - The commutator acts on the projector: $[\hat{r}_\alpha, \beta_I^{lm}]$ is computed by AppNls_0xyz() in the code - Extended algebra in Appendix A - Ref: Mattiat & Luber (2022) Section 2.2, eq.12; Pickard & Mauri (2001)

VERIFICATION NEEDED: trace the sign/i-factor from theory to code. VecPHmatrix() stores $\mathbf{I}_t * \text{block_matrix}$ ($\mathbf{I}_t = i$) in Pxyz. CurrentNlpp() adds $\mathbf{I}_t * \text{block_matrix}$. Current extracted as $\text{Re}(\text{zdotc}(\mathbf{P}, \mathbf{P}_x))$. The -i from the gradient operator + i from storage = real observable. Verify this chain explicitly in the derivation. See Gap 1 in CONTEXT.md.

The velocity operator for a system described by a Hamiltonian containing a nonlocal pseudopotential V^{nl} is obtained from the Heisenberg equation of motion,

$$\hat{v}_\alpha = \frac{1}{i}[\hat{r}_\alpha, \hat{H}] = \hat{p}_\alpha + \frac{1}{i}[\hat{r}_\alpha, V^{\text{nl}}], \quad (41)$$

where the second term on the right-hand side arises because $[\hat{r}_\alpha, V^{\text{nl}}] \neq 0$ for a nonlocal operator. ? ? [Expand into full derivation of current operator, paramagnetic and diamagnetic decomposition, density matrix formulation, and explicit KB pseudopotential commutator. Follow derivation philosophy: complete first, then decide what moves to appendix.]

The macroscopic current density is expressed in terms of the one-particle density matrix $\mathbf{P}(t)$ as

$$J_\alpha(t) = \text{Re Tr}[\mathbf{P}(t) \cdot \mathbf{J}^\alpha], \quad (42)$$

where $\mathbf{J}^\alpha$ is the matrix of current operator elements in the Kohn-Sham orbital basis,

$$J_{ab}^\alpha = \langle \psi_a | \left(-i\nabla_\alpha + k_\alpha + \frac{1}{i}[\hat{r}_\alpha, V^{\text{nl}}] \right) | \psi_b \rangle. \quad (43)$$

[The k_α term in eq. 43 arises from the Bloch theorem: for a periodic orbital $\psi_{n,\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{n,\mathbf{k}}(\mathbf{r})$, the gradient operator acting on the full Bloch function gives $\nabla(e^{i\mathbf{k}\cdot\mathbf{r}}u) = e^{i\mathbf{k}\cdot\mathbf{r}}(\nabla + i\mathbf{k})u$. Derive this step explicitly.]

8.6 Density Matrix Propagation

[Adapt from paper 1 (Jakowski et al. 2025, Section 2.2).] The derivation is identical to paper 1 for the case $S=1$ (orthogonal basis at Gamma point). For $k \neq \text{Gamma}$, H is complex-valued (Hermitian but not real-symmetric) and Ω is also complex. The BCH expansion is the same but with complex matrix arithmetic. Key additions vs paper 1: - Note explicitly that for $k \neq \text{Gamma}$ the Hamiltonian matrix $\mathbf{H}_{\mathbf{k}} = \langle \psi_a | \hat{H} | \psi_b \rangle$ is complex Hermitian - The BCH propagator (Ieldyn=1) handles complex matrices; the diagonalization path (Ieldyn=2) does NOT – state this - The self-consistency loop (predictor-corrector) is unchanged - The Hamiltonian at each time step includes the $A(t)*p$ coupling term (for VECTOR_POT mode); in the perturbative kick scheme A appears only in the initial H_0 setup Reference sections from paper 1: Magnus expansion and density matrix propagation subsections. Cite: Jakowski & Morokuma (2009); Jakowski et al. JCTC 2025.

The time evolution of the one-particle density matrix $\mathbf{P}(t)$ is governed by the Liouville-von Neumann equation

$$\frac{\partial \mathbf{P}(t)}{\partial t} = -i [\mathbf{H}(t), \mathbf{P}(t)], \quad (44)$$

where $\mathbf{H}(t)$ is the Kohn-Sham Hamiltonian matrix in the basis of ground-state Kohn-Sham orbitals. [Continue with Magnus expansion (refer to paper 1 eqs. and cite), then commutator expansion for $\mathbf{P}(t+dt)$, self-consistency loop. For complex H : add note about BCH-only path.]

[Note from code inspection (CONTEXT.md Gap 4): `magnus()` in the code is documented as real-only, but for $k \neq \Gamma$ the call site casts to `double*`. This is mathematically correct since the Magnus operation $\Omega = -i \cdot \frac{1}{2}(H_0 + H_1)\Delta t$ is element-wise and valid for both real and complex H , but should be stated explicitly in the text.]

8.7 Spin-Polarized Extension

Content outline: - Collinear spin-polarized DFT: two independent sets of KS orbitals $\{\psi_n^\uparrow\}$ and $\{\psi_n^\downarrow\}$, each with its own density matrix $\mathbf{P}^\uparrow(t)$ and $\mathbf{P}^\downarrow(t)$ - Spin-resolved electron density: $\rho^\uparrow(\mathbf{r}, t)$ and $\rho^\downarrow(\mathbf{r}, t)$ - Total density: $\rho = \rho^\uparrow + \rho^\downarrow$ - Spin magnetization: $m_s = \rho^\uparrow - \rho^\downarrow$ - Coupling between spin channels: through the spin-dependent XC potential $V_{\text{xc}}^\sigma[\rho^\uparrow, \rho^\downarrow]$ (LSDA or GGA spin-polarized functional) - Each spin channel propagates independently: $\partial \mathbf{P}^\sigma / \partial t = -i[\mathbf{H}^\sigma(t), \mathbf{P}^\sigma(t)]$ - The Hamiltonians $\mathbf{H}^\uparrow$ and $\mathbf{H}^\downarrow$ differ only through $V_{\text{xc}}^\uparrow \neq V_{\text{xc}}^\downarrow$ - Spin-resolved current and optical conductivity: $J_\alpha^\sigma(t) = \text{Re Tr}[\mathbf{P}^\sigma(t) \cdot \mathbf{J}^\alpha] \sigma^\sigma(\omega)$ for each spin channel - Total and spin currents: $\mathbf{J}_{\text{charge}} = \mathbf{J}^\uparrow + \mathbf{J}^\downarrow$, $\mathbf{J}_{\text{spin}} = \mathbf{J}^\uparrow - \mathbf{J}^\downarrow$ - Note: non-collinear spin (two-component

spinors) is beyond the scope of this work and will be presented in a future publication. - Note from code inspection: spin is handled via separate MPI communicator groups (pct.spinpe); spin channels communicate only when updating V_{xc} (rho.get_oppo()).

In the collinear spin-polarized extension, the density matrix propagation (Section 8.6) is performed independently for each spin channel $\sigma \in \{\uparrow, \downarrow\}$,

$$\frac{\partial \mathbf{P}^\sigma(t)}{\partial t} = -i [\mathbf{H}^\sigma(t), \mathbf{P}^\sigma(t)], \quad (45)$$

where the spin-dependent Hamiltonian $\mathbf{H}^\sigma(t)$ contains the spin-resolved exchange-correlation potential $V_{xc}^\sigma[\rho^\uparrow(\mathbf{r}, t), \rho^\downarrow(\mathbf{r}, t)]$. [Continue: derive spin-resolved current, define total and spin currents, note that $J_spin = J_up - J_down$ is available as a low-cost observable.]

8.8 Postprocessing: Optical Conductivity and Berry Phase

(a) From $J(t)$ to optical conductivity: - Fourier transform: $\tilde{J}_\alpha(\omega) = \int J_\alpha(t) e^{i\omega t} dt$ - Optical conductivity tensor: $\sigma_{\alpha\beta}(\omega) = \tilde{J}_\alpha(\omega)/E_\beta(\omega)$ - Dielectric function: $\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + i\sigma_{\alpha\beta}(\omega)/\varepsilon_0\omega$ - In practice: apply delta-kick perturbation in direction β , Fourier transform $J_\alpha(t)$, divide by $E_\beta(\omega) = E_0$ (constant for delta kick) - Mention damping/broadening: apply $e^{-\gamma t}$ window before FFT

(b) Berry phase polarization as alternative observable: - For periodic systems, polarization can also be extracted via the King-Smith/Vanderbilt Berry phase formula:[?] $\mathbf{P}_{BP} = \frac{e}{(2\pi)^3} \int_{BZ} \mathbf{A}_n(\mathbf{k}) d\mathbf{k}$ where $\mathbf{A}_n = i\langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} | u_{n\mathbf{k}} \rangle$ is the Berry connection - For the Gamma-point supercell limit, the discrete formula (Resta 1998, eq. 48 in Mattiat & Luber): $\langle \hat{d}_\alpha \rangle = \frac{eL}{2\pi} \text{Im} \ln \det S_\alpha$ where $S_{\alpha,ij} = \langle \psi_i | e^{-2\pi i r_\alpha/L} | \psi_j \rangle$ - In density matrix form: $P_{BP,\alpha}(t) = \text{Tr}[\mathbf{P}(t) \cdot \mathbf{X}^\alpha]$ where $\mathbf{X}^\alpha$ is the Berry phase position matrix - Equivalence with current-integration result: $\Delta P_\alpha = \int_0^t J_\alpha(t') dt'$ (test of internal consistency) - Implementation in RMG: BP_Xml matrix computed by Rmg_BP->tddft_Xml() (see code, RmgTddft.cpp lines 514,517) - Reference: King-Smith & Vanderbilt (1993); Resta (1998); Mattiat & Luber (2022) Section 2.5

The macroscopic current density $\mathbf{J}(t)$ obtained from eq. (42) provides direct access to the optical response of the periodic system. After applying a momentum kick of amplitude E_0 along the Cartesian direction β at $t = 0$, the optical conductivity tensor is obtained from the Fourier transform of the current density as

$$\sigma_{\alpha\beta}(\omega) = \frac{\tilde{J}_\alpha(\omega)}{E_\beta(\omega)} = \frac{1}{E_0} \int_0^\infty J_\alpha(t) e^{i\omega t - \gamma t} dt, \quad (46)$$

where γ is a broadening parameter.[?] [Expand: relate σ to ε ; describe how three independent kick simulations (x, y, z) give the full tensor; note crystal symmetry reduces the number of independent components.]

[Add: Berry phase derivation and density matrix form. Show equivalence with current integration. Note that both observables are computed in the present implementation and their agreement serves as an internal consistency check.]

8.9 Interaction of Matter with Electromagnetic Radiation

The starting point for any RT-TDDFT calculation is the coupling of the electronic system to an external time-dependent perturbation. This section establishes the general framework for the interaction of electrons with electromagnetic radiation, motivates the specific choices of gauge and approximation used in this work, and identifies the observables through which spectral properties are extracted from the simulation.

8.9.1 The general time-dependent Kohn-Sham equation with electromagnetic coupling

The starting point for RT-TDDFT in the presence of an electromagnetic field is the time-dependent Kohn-Sham (TDKS) equation with the standard minimal coupling prescription. For a single KS orbital ψ of an electron (charge $q_e = -e$, with $e > 0$), the TDKS equation reads

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \hat{H}(t) \psi(\mathbf{r}, t) \quad (47)$$

with the Hamiltonian

$$\hat{H}(t) = \frac{1}{2m} \left(\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(\mathbf{r}, t) \right)^2 + q_e \Phi(\mathbf{r}, t) + V_{\text{eff}}[\rho](\mathbf{r}, t). \quad (48)$$

Here $\mathbf{A}(\mathbf{r}, t)$ and $\Phi(\mathbf{r}, t)$ are the vector and scalar potentials describing the electromagnetic field, and $V_{\text{eff}}[\rho]$ is the Kohn-Sham effective potential containing the ionic, Hartree, and exchange-correlation contributions.

In the general form, two electromagnetic potentials ($\mathbf{A}$ and Φ) and the KS potential all appear simultaneously. However, the gauge freedom of electrodynamics—the fact that $\mathbf{A}$ and Φ can be transformed without changing the physical fields $\mathbf{E}$ and $\mathbf{B}$ —allows us to simplify significantly. The choice of gauge determines how the EM coupling enters the Hamiltonian.

Throughout this work we adopt the **Coulomb gauge**, defined by two conditions:

$$\nabla \cdot \mathbf{A} = 0 \quad (\text{transversality}), \quad \Phi_{\text{rad}} = 0 \quad (\text{no scalar potential from radiation}). \quad (49)$$

In this gauge, all information about the radiation field is carried exclusively by $\mathbf{A}(\mathbf{r}, t)$. The only scalar potential that remains is the instantaneous Coulomb interaction between charges, which is already contained in V_{eff} . The TDKS Hamiltonian simplifies to

$$\hat{H}(t) = \frac{1}{2m} \left(\hat{\mathbf{p}} + \frac{e}{c} \mathbf{A}(\mathbf{r}, t) \right)^2 + V_{\text{eff}}[\rho](\mathbf{r}, t), \quad (50)$$

where the sign flip ($-q_e \rightarrow +e$) comes from $q_e = -e$ for the electron.

Expanding the squared kinetic term gives three physically distinct contributions:

$$\hat{H}(t) = \underbrace{\frac{\hat{p}^2}{2m}}_{T_0} + \underbrace{\frac{e}{mc} \mathbf{A} \cdot \hat{\mathbf{p}}}_{\text{paramagnetic}} + \underbrace{\frac{e^2}{2mc^2} \mathbf{A}^2}_{\text{diamagnetic}} + V_{\text{eff}}. \quad (51)$$

The *paramagnetic term* is the leading-order light-matter coupling; it is linear in $\mathbf{A}$ and

couples the field to the electronic current. The *diamagnetic term* is quadratic in $\mathbf{A}$ and is typically negligible for weak perturbative fields (it enters at second order in perturbation theory), though it is required for gauge invariance of the response and for satisfaction of the f-sum rule. In the Coulomb gauge, $\nabla \cdot \mathbf{A} = 0$ guarantees that $\mathbf{A} \cdot \hat{\mathbf{p}} = \hat{\mathbf{p}} \cdot \mathbf{A}$, so the paramagnetic term simplifies to a single $(e/mc)\mathbf{A} \cdot \hat{\mathbf{p}}$ without symmetrization.

[\[Molecular analogy:\]](#) In Gaussian basis set codes, the same Hamiltonian appears but $\hat{\mathbf{p}}$ is represented as matrix elements $\langle \phi_\mu | \hat{\mathbf{p}} | \phi_\nu \rangle$ between atomic orbitals. In the real-space grid representation used by RMG, $\hat{\mathbf{p}} = -i\hbar\nabla$ acts directly through finite-difference stencils on the grid—no basis transformation is needed.

8.9.2 The electromagnetic plane wave

A monochromatic EM plane wave in the Coulomb gauge is described by the vector potential

$$\mathbf{A}(\mathbf{r}, t) = A_0 \hat{\varepsilon} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} + \text{c.c.}, \quad (52)$$

where $\hat{\varepsilon}$ is the polarization unit vector, $\mathbf{q}$ is the photon wavevector with $|\mathbf{q}| = \omega/c$, and ω is the angular frequency. The transversality condition $\nabla \cdot \mathbf{A} = 0$ requires $\hat{\varepsilon} \cdot \mathbf{q} = 0$: the polarization is perpendicular to the direction of propagation.

The physical electric and magnetic fields are obtained from $\mathbf{A}$ alone through Maxwell’s relations:

$$\mathbf{E}(\mathbf{r}, t) = -\frac{1}{c} \frac{\partial \mathbf{A}}{\partial t} = \frac{i\omega A_0}{c} \hat{\varepsilon} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} + \text{c.c.}, \quad (53)$$

$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A} = iA_0 (\mathbf{q} \times \hat{\varepsilon}) e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} + \text{c.c.} \quad (54)$$

This confirms that a single vector potential in the Coulomb gauge is sufficient to represent both physical fields—no scalar potential is needed. In Gaussian units, $|\mathbf{B}|/|\mathbf{E}| = 1$ for a plane wave, but the coupling of non-relativistic electrons to $\mathbf{B}$ is weaker than to $\mathbf{E}$ by a factor of $v/c \approx \alpha \approx 1/137$. This hierarchy is the physical origin of the dominance of electric-dipole transitions in ordinary spectroscopy.

8.9.3 Multipole expansion and the long-wavelength approximation

The spatial factor $e^{i\mathbf{q} \cdot \mathbf{r}}$ in the vector potential can be expanded in a Taylor series:

$$e^{i\mathbf{q} \cdot \mathbf{r}} = 1 + i\mathbf{q} \cdot \mathbf{r} - \frac{1}{2}(\mathbf{q} \cdot \mathbf{r})^2 + \cdots \quad (55)$$

Each term corresponds to a class of multipole transitions with distinct selection rules and characteristic coupling strengths.

The **zeroth-order term** (1) retains only the temporal oscillation of $\mathbf{A}$, making it spatially uniform: $\mathbf{A}(\mathbf{r}, t) \rightarrow \mathbf{A}(t)$. This is the **long-wavelength** or **dipole approximation**. The resulting $(e/mc)\mathbf{A}(t) \cdot \hat{\mathbf{p}}$ interaction drives electric dipole (E1) transitions—the dominant contribution to optical absorption. The approximation is valid when the spatial extent of the electronic system is much smaller than the photon wavelength: $|\mathbf{q}||\mathbf{r}| \ll 1$. For visible light ($\lambda \approx 400\text{--}700\text{ nm}$) interacting with molecules ($\sim 1\text{ nm}$) or even nanoscale systems ($\sim 10\text{ nm}$),

the ratio $|\mathbf{q}||\mathbf{r}| \approx 10^{-2}$ – 10^{-1} is small and the approximation is excellent.

The **first-order term** ($i\mathbf{q} \cdot \mathbf{r}$) gives the electric quadrupole (E2) and magnetic dipole (M1) contributions. These are suppressed by a factor of $|\mathbf{q}||\mathbf{r}| \approx v/c$ relative to E1 and are important for forbidden transitions, circular dichroism, and X-ray spectroscopy where λ approaches atomic dimensions.

Higher-order terms give electric octupole, magnetic quadrupole, etc., relevant only for very short wavelengths or very extended systems.

The long-wavelength approximation (retaining only the zeroth-order term) has been the standard in molecular quantum chemistry, including in paper 1 of this series.[?] However, it has well-known limitations that become critical for periodic systems, as discussed next.

8.9.4 Failure of the length gauge for periodic systems

In the long-wavelength limit with spatially uniform $\mathbf{A}(t)$, two equivalent representations of the light-matter interaction are available, connected by the Göppert-Mayer gauge transformation (derived in Section ??):

Velocity gauge (VG). The interaction enters through the kinetic energy as $(e/mc)\mathbf{A}(t) \cdot \hat{\mathbf{p}}$. The perturbation operator involves $\hat{\mathbf{p}} = -i\hbar\nabla$, which is well-defined for any basis set or boundary condition.

Length gauge (LG). A gauge transformation with $\chi(\mathbf{r}, t) = -\mathbf{r} \cdot \mathbf{A}(t)$ eliminates $\mathbf{A}$ from the kinetic energy and transfers the interaction to a scalar potential $\Phi = \mathbf{E}(t) \cdot \mathbf{r}$, giving the familiar dipole interaction $-e\mathbf{E}(t) \cdot \hat{\mathbf{r}}$. The perturbation operator now involves the position operator $\hat{\mathbf{r}}$.

Note that these two gauges both operate within the Coulomb gauge framework—they are different representations of the same physics obtained by choosing different gauge functions, not different physical theories. For *finite* molecular systems, both give identical results, and the length gauge is often preferred because the dipole matrix elements $\langle\psi_a|\hat{r}_\alpha|\psi_b\rangle$ have a transparent physical interpretation as transition dipole moments. This was the approach used in paper 1.[?]

For *periodic* systems under Born-von Kármán boundary conditions, the length gauge fails catastrophically. The position operator $\hat{\mathbf{r}}$ is fundamentally incompatible with periodicity: $\mathbf{r}$ and $\mathbf{r} + \mathbf{L}$ must be physically equivalent, but $\hat{\mathbf{r}}$ distinguishes them. The scalar potential $\mathbf{E} \cdot \mathbf{r}$ is a linearly growing ramp that produces a discontinuous “sawtooth” at the unit cell boundary—it is not periodic and cannot be represented in a Bloch basis. The matrix elements $\langle\psi_{n\mathbf{k}}|\hat{r}_\alpha|\psi_{m\mathbf{k}}\rangle$ between Bloch states are ill-defined (this is precisely the problem resolved by the modern theory of polarization via Berry phases, discussed in Section 8.9.7).

The velocity gauge completely avoids this problem: the momentum operator $\hat{\mathbf{p}}$ is perfectly well-defined for Bloch states, and the matrix elements $\langle\psi_{n\mathbf{k}}|\hat{p}_\alpha|\psi_{m\mathbf{k}}\rangle$ exist without any difficulty. This is why periodic-system RT-TDDFT implementations, including the present one, use the velocity gauge.

8.9.5 Scope and limitations of the velocity gauge with the dipole approximation

The velocity gauge resolves the periodic-boundary problem, but the underlying long-wavelength approximation itself imposes a fundamental limitation: it restricts the calculation to **vertical transitions** with zero crystal momentum transfer ($\Delta\mathbf{k} = 0$). This means that only the optical limit of the dielectric function $\varepsilon(\mathbf{q} \rightarrow 0, \omega)$ is accessible.

It is sometimes stated that the velocity gauge “is not general,” but this is imprecise. The gauge transformation itself is exact—gauge transformations are always unitary and introduce no approximation. What is limited is the *dipole approximation* that precedes it: by setting $e^{i\mathbf{q}\cdot\mathbf{r}} \rightarrow 1$, one discards all finite- $\mathbf{q}$ physics regardless of which gauge is used. The velocity gauge is the correct and rigorous framework for $q = 0$ optical response in periodic systems; the limitation lies in the $q = 0$ restriction, not in the gauge.

Many important spectroscopies and collective excitations involve finite momentum transfer and lie beyond this approximation. These include plasmon dispersion $\varepsilon(\mathbf{q}, \omega)$, electron energy-loss spectroscopy (EELS) where the beam transfers finite $\mathbf{q}$ to the sample, inelastic X-ray scattering (IXS), and magnon dispersion in magnetic systems.

8.9.6 Beyond the dipole approximation: the full vector potential

A natural generalization is to retain the full spatial dependence of the vector potential without making the dipole approximation:

$$\hat{H}(t) = \frac{1}{2m} \left(\hat{\mathbf{p}} + \frac{e}{c} \mathbf{A}(\mathbf{r}, t) \right)^2 + V_{\text{eff}}[\rho](\mathbf{r}, t) \quad (56)$$

with $\mathbf{A}(\mathbf{r}, t) = A_0 \hat{\varepsilon} e^{i(\mathbf{q}\cdot\mathbf{r} - \omega t)} + \text{c.c.}$ retaining the full plane-wave spatial structure. This is still the Coulomb gauge—no new gauge choice is needed. The spatial factor $e^{i\mathbf{q}\cdot\mathbf{r}}$ is automatically periodic (it is a reciprocal-space function commensurate with the lattice for appropriate $\mathbf{q}$), so the problematic $\hat{\mathbf{r}}$ operator never appears. Transitions now satisfy crystal momentum conservation $\mathbf{k}_f = \mathbf{k}_i + \mathbf{q} \pmod{\mathbf{G}}$, allowing finite- $\mathbf{q}$ excitations.

This approach is more general than the dipole approximation because the full exponential automatically includes all multipole orders. For finite but small $|\mathbf{q}||\mathbf{r}|$, the leading correction beyond E1 is the electric quadrupole / magnetic dipole term (first order in $\mathbf{q} \cdot \mathbf{r}$). For $|\mathbf{q}||\mathbf{r}| \sim 1$ (as in X-ray spectroscopy or large nanostructures), all orders contribute and the full exponential must be retained.

Important distinction regarding finite- $\mathbf{q}$ response. There is a second, independent route to finite- $\mathbf{q}$ excitations that does not involve the vector potential at all. One can apply a scalar potential perturbation

$$\delta V(\mathbf{r}, t) = V_0 e^{i\mathbf{q}\cdot\mathbf{r}} \delta(t) \quad (57)$$

directly to V_{eff} , with no gauge considerations. This “density-wave kick” probes the density-density response function $\chi(\mathbf{q}, \omega)$, whereas the EM route probes the current-current response $\sigma(\mathbf{q}, \omega)$. The two response functions are related but distinct: σ is a tensor (3×3), χ is a scalar; σ naturally gives the transverse (optical) response, χ gives the longitudinal response relevant

to EELS. For the longitudinal response at finite $\mathbf{q}$, the scalar perturbation approach is more direct. For transverse (optical) response including finite- $\mathbf{q}$ corrections (beyond dipole), the full $\mathbf{A}(\mathbf{r}, t)$ approach is the natural choice.

The present implementation uses the long-wavelength limit (spatially uniform $\mathbf{A}(t)$) exclusively. The general framework described here indicates the natural extension path: keeping the full $e^{i\mathbf{q}\cdot\mathbf{r}}$ spatial structure in $\mathbf{A}$ for beyond-dipole optical response, or using scalar density-wave perturbations for longitudinal excitations at arbitrary $\mathbf{q}$. Such extensions would enable simulation of plasmon dispersion, EELS spectra, inelastic X-ray scattering, and magnon dispersion from first principles—all within the same RT-TDDFT framework.

8.9.7 Observables: from dipole moment to current density to Berry phase polarization

The choice of gauge determines not only how the perturbation enters the Hamiltonian but also which observable provides the most natural route to spectral properties. Three complementary approaches exist, corresponding to three equivalent but operationally distinct ways of characterizing the electronic response.

Length gauge—dipole moment. In the length gauge, the EM field couples through the scalar potential $\mathbf{E}(t) \cdot \hat{\mathbf{r}}$, acting directly on the charge density. The natural observable is the time-dependent dipole moment:

$$\mu_\alpha(t) = -e \sum_n f_n \langle \psi_n(t) | \hat{r}_\alpha | \psi_n(t) \rangle, \quad (58)$$

where the sum runs over occupied KS orbitals with occupations f_n . The Fourier transform of $\mu(t)$ after an impulsive perturbation yields the frequency-dependent polarizability tensor $\alpha(\omega)$, from which the photoabsorption cross section follows as $\sigma_{\text{abs}}(\omega) = (4\pi\omega/c) \text{Im } \alpha(\omega)$. For finite molecular systems this is the standard RT-TDDFT post-processing route, and it was used in paper 1.[?]

Higher-order multipole moments—the quadrupole $Q_{\alpha\beta}(t) = \langle r_\alpha r_\beta \rangle$, the octupole, etc.—can be extracted analogously and provide access to E2, M1, and higher-order contributions to optical activity and circular dichroism, though these are rarely needed for systems small compared to the photon wavelength.

Velocity gauge—electronic current density. In the velocity gauge, the EM field couples through $\mathbf{A}(t) \cdot \hat{\mathbf{p}}$, acting on the electronic momentum. The natural observable is the time-dependent electronic current density:

$$\mathbf{j}(\mathbf{r}, t) = \frac{1}{2} \sum_n f_n [\psi_n^*(\mathbf{r}, t) \hat{\mathbf{p}} \psi_n(\mathbf{r}, t) + \text{c.c.}] + \mathbf{j}_{\text{NL}}(\mathbf{r}, t). \quad (59)$$

The first term is the familiar paramagnetic current—the same quantity that appears in probability current expressions in introductory quantum mechanics. The second term, $\mathbf{j}_{\text{NL}}$, arises from the nonlocal pseudopotential: because the Kleinman-Bylander projectors do not commute with the position operator, the continuity equation $\partial\rho/\partial t + \nabla \cdot \mathbf{j} = 0$ is satisfied

only when the nonlocal contribution is included. The derivation and explicit form of $\mathbf{j}_{\text{NL}}$ are given in Section 8.5. [Verify cross-reference label for Sec. 2.5.]

The macroscopic (cell-averaged) current $\mathbf{J}(t) = (1/\Omega) \int_{\text{cell}} \mathbf{j}(\mathbf{r}, t) d^3r$ connects to optical properties through the frequency-dependent optical conductivity:

$$\sigma_{\alpha\beta}(\omega) = \frac{J_{\alpha}(\omega)}{E_{\beta}(\omega)}, \quad (60)$$

and the dielectric function:

$$\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + \frac{4\pi i}{\omega} \sigma_{\alpha\beta}(\omega), \quad (61)$$

from which absorption spectra, reflectivity, refractive index, and energy-loss functions follow by standard relations. This is the periodic-system analog of the molecular polarizability route: where molecular RT-TDDFT extracts $\alpha(\omega)$ from $\mu(t)$, periodic RT-TDDFT extracts $\varepsilon(\omega)$ from $\mathbf{J}(t)$.

Berry phase—macroscopic polarization. The macroscopic polarization $\mathbf{P}$ of a periodic system cannot be defined as the expectation value of any local operator—this is the same fundamental difficulty with $\hat{\mathbf{r}}$ discussed in Section 8.9.4. The modern theory of polarization^{7,8} resolves this by expressing $\mathbf{P}$ as a geometric phase (Berry phase) of the occupied Bloch orbitals:

$$P_{\alpha} = \frac{-e}{(2\pi)^3} \sum_n \int_{\text{BZ}} d\mathbf{k} \langle u_{n\mathbf{k}} | i \frac{\partial}{\partial k_{\alpha}} | u_{n\mathbf{k}} \rangle, \quad (62)$$

where $u_{n\mathbf{k}}(\mathbf{r})$ is the cell-periodic part of the Bloch orbital. This expression is defined modulo a quantum of polarization $e\mathbf{R}/\Omega$ (where $\mathbf{R}$ is a lattice vector and Ω is the cell volume), reflecting the intrinsic multivaluedness of the geometric phase.

The Berry phase polarization is the periodic-system generalization of the molecular dipole moment. Just as $\mu = -e\langle \hat{\mathbf{r}} \rangle$ is a single well-defined number for a localized molecule, P^{Berry} is a well-defined number (modulo the quantum) for an infinite periodic crystal. The Berry phase accomplishes for periodic systems what the position operator does for molecules—but through a topological construction in k -space rather than a real-space expectation value.

In RT-TDDFT, the time-evolved Bloch orbitals $\psi_{n\mathbf{k}}(t)$ at each time step define a time-dependent Berry phase, giving the polarization $\mathbf{P}(t)$ directly without reference to the position operator. The connection to the current is exact:

$$\mathbf{J}(t) = \frac{d\mathbf{P}}{dt}, \quad (63)$$

so the polarization can equivalently be recovered by time-integration of the current:

$$\mathbf{P}(t) = \mathbf{P}(0) + \int_0^t \mathbf{J}(t') dt'. \quad (64)$$

The two routes—direct Berry phase evaluation at each step versus current integration—are formally equivalent but have different numerical characteristics. The Berry phase gives $\mathbf{P}(t)$ without accumulating integration error, but requires dense k -point sampling (the formula

is a k -space integral) and careful tracking of the polarization quantum for large-amplitude perturbations. Current integration avoids the Berry phase calculation but accumulates time-stepping error over long propagations. In the linear response regime (small perturbations), both approaches are reliable and serve as mutual consistency checks. The Berry phase formalism and its implementation in the present code are described in detail in Section ??.

[Verify cross-reference label for Sec. 2.8.]

Equivalence, complementarity, and the choice of observable. For finite systems where $\hat{\mathbf{r}}$, $\hat{\mathbf{p}}$, and the Berry phase are all well-defined, the three routes are exactly equivalent, connected by the time derivative of the dipole moment:

$$\frac{d\mu_\alpha}{dt} = -e \sum_n f_n \langle \psi_n(t) | \hat{v}_\alpha | \psi_n(t) \rangle = \Omega J_\alpha(t) = \Omega \frac{dP_\alpha}{dt}. \quad (65)$$

In frequency space this yields the fundamental relation $\sigma(\omega) = i\omega \alpha(\omega)$, linking the polarizability to the conductivity, and the dielectric function $\varepsilon(\omega) = 1 + 4\pi\alpha(\omega)/\Omega = 1 + 4\pi i\sigma(\omega)/\omega$.

For periodic systems, the dipole moment is undefined, leaving two viable routes: the current density $\mathbf{J}(t)$, which gives $\sigma(\omega)$ and $\varepsilon(\omega)$ directly via Fourier transform; and the Berry phase $\mathbf{P}(t)$, which gives $\alpha(\omega)$ and connects to the absolute polarization. In the present work, we use the current density as the primary observable (Sections 8.5–[2.6]) and the Berry phase as a complementary diagnostic and for properties where the absolute polarization is needed (Section ??).

There is an appealing conceptual symmetry in the gauge-observable pairing: in the length gauge, the field perturbs the density (through $\mathbf{E} \cdot \hat{\mathbf{r}}$) and the response is read from the density’s moments; in the velocity gauge, the field perturbs the current (through $\mathbf{A} \cdot \hat{\mathbf{p}}$) and the response is read from the current. The Berry phase stands apart from this pattern—it is a gauge-invariant geometric property of the occupied manifold that can be evaluated regardless of how the perturbation was applied, depending only on the instantaneous state of the system, not on the history of how it was driven.

8.9.8 Summary and scope of the present work

The general TDKS equation with full minimal coupling to $\mathbf{A}(\mathbf{r}, t)$ in the Coulomb gauge provides a unified framework for modeling the interaction of electronic systems with electromagnetic radiation. The key approximations adopted in the present work and their consequences are as follows.

The **Coulomb gauge** ($\nabla \cdot \mathbf{A} = 0$, $\Phi_{\text{rad}} = 0$) carries all radiation physics in $\mathbf{A}$ alone, avoiding ambiguity between scalar and vector potential contributions. Both electric and magnetic fields are derivable from $\mathbf{A}$ via standard Maxwell relations.

The **long-wavelength approximation** ($e^{i\mathbf{q} \cdot \mathbf{r}} \rightarrow 1$) reduces $\mathbf{A}(\mathbf{r}, t)$ to a spatially uniform $\mathbf{A}(t)$, restricting the calculation to vertical transitions ($\Delta\mathbf{k} = 0$). This is equivalent to the electric dipole approximation and is valid when the photon wavelength far exceeds the system size.

The **velocity gauge** represents the interaction as $(e/mc)\mathbf{A}(t) \cdot \hat{\mathbf{p}}$ in the kinetic energy, avoiding the ill-defined position operator that plagues the length gauge in periodic systems.

The **current density** is the primary observable, giving the optical conductivity $\sigma(\omega)$ and dielectric function $\varepsilon(\omega)$ via Fourier transform. The **Berry phase polarization** provides a complementary route to the same spectral information and additionally gives the absolute macroscopic polarization.

The present work operates within all three approximations: Coulomb gauge, long-wavelength limit, velocity gauge. This gives access to the full optical response $\varepsilon(\omega)$ of periodic systems with correct treatment of local field effects, current contributions from nonlocal pseudopotentials, and spin-collinear dynamics. Extensions to finite- $\mathbf{q}$ response—via either the full $\mathbf{A}(\mathbf{r}, t)$ or scalar density-wave perturbations—and to higher-multipole processes are natural next steps that the framework accommodates without conceptual difficulty.

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A Gauge Transformation of the Nonlocal Pseudopotential Term

[Full derivation of the gauge transformation of V^{nl} from length to velocity gauge, following the structure of Mattiat & Luber (2022) Appendix B but adapted to the Kleinman-Bylander form used in RMG. This is the extended algebra supporting section 8 subsection on current operator and nonlocal pseudopotentials.]

B Equivalence of Current-Based and Berry Phase Polarization

[Proof that the velocity gauge current $\mathbf{J}(t)$, when integrated, gives the same polarization as the King-Smith/Vanderbilt Berry phase formula. This supports the comparison presented in section ??.]

C [Additional appendix if needed]

[Placeholder. May be used for: spin-collinear extension of the BCH scheme; extended orthogonalization proofs if needed beyond paper 1 appendix; or can be removed if not needed.]

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